



AZ-900

Microsoft Azure Fundamentals

Interview Questions & Answers

Complete Azure Fundamentals Preparation Guide

2025 EDITION

300

Questions

10

Topics

Beginner

to Advanced

Author: Akhilesh Chaudhari

Microsoft Azure Certification Preparation Guide

AZ-900 Certification Preparation | 2025 Edition

TABLE OF CONTENTS

AZ-900 Interview Questions & Answers — 300 Questions | 10 Sections

Preface & Introduction

3

1

Cloud Concepts

Q1 – Q30

4

2

Azure Core Services — Compute

Q31 – Q60

5

3

Azure Core Services — Storage

Q61 – Q90

6

4

Azure Core Services — Networking

Q91 – Q120

7

5

Azure Identity, Access & Security

Q121 – Q150

8

6

Azure Management & Governance

Q151 – Q180

9

7

Azure AI, ML & Modern Services

Q181 – Q210

10

8

Azure Databases

Q211 – Q240

11

9

Azure DevOps & Developer Tools

Q241 – Q270

12

10

Azure Pricing, Support & Exam Prep

Q271 – Q300

13

300 Questions

10 Sections

Beginner to Advanced

AZ-900 | 2025 Edition

PREFACE & INTRODUCTION

Welcome to the AZ-900: Microsoft Azure Fundamentals Interview Questions & Answers guide — your comprehensive companion for Microsoft Azure certification preparation.

The AZ-900 exam validates foundational knowledge of cloud services and how those services are provided with Microsoft Azure. This guide contains 300 carefully crafted interview questions and answers spanning all major AZ-900 exam domains — from cloud concepts to Azure pricing, governance, and compliance.

Topics Covered: Cloud Concepts • Azure Compute • Azure Storage • Azure Networking • Identity & Security • Management & Governance • AI & Modern Services • Azure Databases • Azure DevOps & Developer Tools • Pricing, Support & Exam Prep

How to use this guide: Read through questions progressively. Try to answer each question before reading the provided answer. Focus on understanding concepts — AZ-900 tests conceptual understanding, not deep technical implementation.

Best of luck in your AZ-900 certification journey!

— Akhilesh Chaudhari

Section 1: Cloud Concepts

Questions 1 – 30

Q1 What is cloud computing?

Answer : Cloud computing is the delivery of computing services — including servers, storage, databases, networking, software, analytics, and intelligence — over the internet ('the cloud') to offer faster innovation, flexible resources, and economies of scale. Instead of owning physical hardware, organizations pay only for cloud services they use.

Q2 What are the three main cloud service models?

Answer : IaaS (Infrastructure as a Service): provides virtualized computing resources like VMs, storage, and networking. PaaS (Platform as a Service): provides a platform for developers to build, test, and deploy applications without managing the underlying infrastructure. SaaS (Software as a Service): delivers software applications over the internet on a subscription basis (e.g., Microsoft 365, Outlook).

Q3 What is the difference between CapEx and OpEx in cloud?

Answer : CapEx (Capital Expenditure) is upfront spending on physical infrastructure — buying servers, data centers. It has high initial cost but assets depreciate over time. OpEx (Operational Expenditure) is ongoing spending for services — paying for cloud resources as you use them. Cloud shifts IT spending from CapEx to OpEx, reducing large upfront investments and allowing pay-as-you-go pricing.

Q4 What are the main cloud deployment models?

Answer : Public Cloud: resources owned and operated by a third-party provider (Microsoft Azure, AWS, Google Cloud) and shared among multiple tenants. Private Cloud: resources used exclusively by one organization, hosted on-premises or by a provider. Hybrid Cloud: combines public and private clouds, allowing data and applications to move between them for greater flexibility and optimization.

Q5 What is the consumption-based model?

Answer : The consumption-based model means customers only pay for what they use, with no upfront costs and no wasted resources. Azure charges based on actual resource usage (CPU time, storage consumed, data transferred). This allows organizations to scale costs with demand, reducing expenses during low usage periods and scaling up during peak times.

Q6 What are the benefits of high availability in cloud?

Answer : High availability ensures services remain accessible with minimal downtime. Azure offers SLAs (Service Level Agreements) guaranteeing uptime percentages (e.g., 99.9% or 99.99%). This is achieved through redundant systems, failover mechanisms, load balancing, and geographic distribution. High availability is critical for business-critical applications that cannot tolerate interruptions.

Q7 What is scalability in cloud computing?

Answer : Scalability is the ability to increase or decrease resources to meet demand. Vertical scaling (scale up/down): adding more CPU or RAM to an existing resource. Horizontal scaling (scale out/in): adding more instances of a resource (e.g., more VMs). Azure supports both models, allowing automatic scaling based on metrics like CPU usage or incoming request volume.

Q8 What is the difference between scalability and elasticity?

Answer : Scalability is the ability to handle increased workload by adding resources (planned growth). Elasticity is the ability to automatically scale resources up or down in real time based on demand fluctuations. Elasticity is dynamic and automatic — Azure can add VMs during a traffic spike and remove them when demand drops, ensuring optimal cost and performance.

Q9 What is agility in cloud computing?

Answer : Agility refers to the ability to rapidly develop, test, and deploy applications in the cloud. Cloud resources can be provisioned in minutes instead of weeks. This enables organizations to quickly respond to changing business requirements, experiment with new ideas, and accelerate time to market for products and services without lengthy procurement processes.

Q10 What is disaster recovery in cloud?

Answer : Disaster recovery (DR) is the strategy for restoring IT systems and data after a catastrophic event (hardware failure, cyberattack, natural disaster). Azure provides tools like Azure Site Recovery, geo-redundant storage, and backup services. DR in cloud is faster and more cost-effective than traditional DR sites because resources are provisioned only when needed.

Q11 What is the shared responsibility model?

Answer : The shared responsibility model defines what Microsoft and the customer are each responsible for securing. Microsoft is always responsible for physical security, network controls, and the host infrastructure. Customer responsibility varies by service type: in IaaS customers manage OS, apps, and data; in PaaS Microsoft manages OS and runtime; in SaaS Microsoft manages everything except data and access.

Q12 What is a Service Level Agreement (SLA) in Azure?

Answer : An SLA is a formal contract between Microsoft and customers defining the guaranteed level of service, primarily uptime and connectivity. Azure SLAs are expressed as percentages (e.g., 99.9% uptime = 8.7 hours downtime/year; 99.99% = 52 minutes/year). Services without SLAs (like free tiers or preview services) have no uptime guarantees.

Q13 What is the Azure global infrastructure?

Answer : Azure's global infrastructure consists of datacenters organized into regions, availability zones, and geographies. Azure has 60+ regions worldwide — more than any other cloud provider. Each region is a set of datacenters connected by a low-latency network. This global footprint enables customers to deploy applications close to their users and meet data residency requirements.

Q14 What is an Azure Region?

Answer : An Azure region is a geographic area containing one or more datacenters connected by a low-latency network. Examples include East US, West Europe, and Southeast Asia. When deploying resources, you choose a region to determine where your data is physically stored. Some services are region-specific while others are global. Region choice affects latency, compliance, and pricing.

Q15 What are Azure Availability Zones?

Answer : Availability Zones are physically separate datacenters within an Azure region, each with independent power, cooling, and networking. They protect applications from datacenter failures. Deploying resources across multiple Availability Zones provides high availability — if one zone fails, others continue operating. Not all regions have Availability Zones. Azure guarantees 99.99% uptime for VMs deployed across zones.

Q16 What is an Azure Region Pair?

Answer : Each Azure region is paired with another region at least 300 miles away within the same geography. Region pairs are used for geo-redundant storage and disaster recovery. During planned maintenance, Azure updates one region in a pair at a time to reduce downtime risk. In a disaster, one region in each pair is prioritized for recovery. Examples: East US paired with West US; UK South with UK West.

Q17 What is a geography in Azure?

Answer : An Azure geography is a discrete market (typically a country or group of countries) that preserves data residency and compliance requirements. Each geography contains at least two Azure regions. Geographies keep customer data within defined boundaries to meet local regulatory requirements (e.g., EU data must stay in Europe). Examples: United States, Europe, Asia Pacific, Brazil.

Q18 What are the advantages of cloud computing?

Answer : Key advantages: High availability and disaster recovery built in. Global reach — deploy anywhere in minutes. Economies of scale — Microsoft's buying power reduces costs. Pay-as-you-go — no wasted capacity. Elasticity — scale automatically with demand. Security — enterprise-grade security managed by Microsoft. Reduced management overhead — no physical hardware maintenance.

Q19 What is the difference between public and private cloud?

Answer : Public cloud: owned and operated by Microsoft, shared infrastructure among multiple customers (multi-tenant), accessed over the internet, lower cost, no maintenance. Private cloud: dedicated infrastructure for one organization, hosted on-premises or in a datacenter, greater control and customization, higher cost, suitable for strict compliance or security requirements that prohibit shared infrastructure.

Q20 What is a hybrid cloud and when should you use it?

Answer : Hybrid cloud combines public cloud (Azure) and private cloud or on-premises infrastructure, connected securely. Use cases: when data must remain on-premises for compliance, when legacy systems can't move to cloud, for burst capacity (run most workloads on-premises, scale to Azure during peaks), and for gradual cloud migration. Azure Arc and Azure Stack enable hybrid scenarios.

Q21 What is fault tolerance in cloud?

Answer : Fault tolerance is the ability of a system to continue operating correctly even when one or more components fail. Azure achieves this through redundancy: multiple copies of data (replication), multiple VMs behind a load balancer, and automatic failover. Unlike high availability (minimizing downtime), fault tolerance aims for zero downtime even during component failures.

Q22 What is predictability in cloud?

Answer : Cloud predictability has two aspects: Performance predictability — consistent, reliable performance through auto-scaling, load balancing, and high-availability architecture. Cost predictability — ability to forecast spending using Azure's pricing calculators, cost management tools, and consumption monitoring. Predictability helps organizations plan and budget effectively without surprise costs or performance degradation.

Q23 What is security in cloud computing?

Answer : Cloud security is a shared responsibility. Azure provides: physical datacenter security, DDoS protection, encryption at rest and in transit, identity and access management (Azure AD), network security groups, Azure Firewall, Security Center, and compliance certifications (ISO, SOC, HIPAA, GDPR). Customers are responsible for data classification, access controls, and application security.

Q24 What is governance in cloud?

Answer : Cloud governance is the set of policies, processes, and tools that ensure cloud resources are used appropriately, compliantly, and cost-effectively. Azure provides Azure Policy (enforce rules), Management Groups (organize subscriptions), Blueprints (deploy compliant environments), and Cost Management. Governance prevents unauthorized resource creation, ensures compliance, and controls spending.

Q25 What is manageability in cloud?

Answer : Cloud manageability refers to how you manage and interact with cloud resources. Management of the cloud: Azure manages hardware, networking, and infrastructure. Management in the cloud: customers can use Azure Portal (GUI), Azure CLI (command line), PowerShell, REST APIs, and Azure Resource Manager (ARM) templates to deploy and manage resources programmatically.

Q26 What is an SLA composite?

Answer : A composite SLA is the combined uptime of multiple services used together. It's calculated by multiplying individual SLAs. Example: if a web app (99.95% SLA) depends on a database (99.99% SLA), the composite SLA is $99.95\% \times 99.99\% = 99.94\%$. This means adding dependencies can lower the overall SLA, which is why architects use redundancy to maintain high availability.

Q27 What are free Azure services?

Answer : Azure offers several free services: Always free — Azure Active Directory (free tier), Azure DevOps (5 users), Azure Functions (1 million executions/month), Blob Storage (5 GB), Cosmos DB (1,000 RU/s), SQL Database (100,000 vCore seconds). 12-month free — VMs (750 hours), Managed Disks (64 GB), SQL Database (250 GB). Plus \$200 credit for 30 days for new accounts.

Q28 What is the Azure free account?

Answer : The Azure free account gives new customers: \$200 credit to use within 30 days on any Azure service, 12 months of popular free services (VMs, storage, databases), and always-free services with monthly limits. After 30 days or credit exhaustion, you're prompted to upgrade to pay-as-you-go. Credit card is required for identity verification but won't be charged automatically.

Q29 What does 'pay-as-you-go' mean in Azure?

Answer : Pay-as-you-go (PAYG) pricing means you pay only for what you use with no upfront commitment. Azure meters resource usage (compute hours, storage GB, data transferred) and bills monthly. You can stop resources at any time to stop being charged. This model eliminates over-provisioning waste and allows cost alignment with actual business activity. Ideal for variable or unpredictable workloads.

Q30 What is TCO (Total Cost of Ownership) in Azure?

Answer : TCO is the complete cost of using a technology solution, including hardware, software, networking, facilities, and staff. Azure's TCO Calculator helps organizations estimate savings by comparing on-premises costs (servers, power, cooling, IT staff, rack space) with Azure cloud costs. Typically organizations save 30-90% by moving to Azure, primarily by eliminating hardware maintenance and datacenter costs.

Section 2: Azure Core Services – Compute

Questions 31 – 60

Q31 What is Azure Virtual Machines (VMs)?

Answer : Azure Virtual Machines (VMs) provide IaaS compute – virtualized servers in the cloud running Windows or Linux. You have full control over the OS, software, and configuration. VMs are ideal for: lifting and shifting on-premises workloads, running custom software, hosting applications requiring specific OS configurations, and dev/test environments. You pay for compute and storage used.

Q32 What are VM Scale Sets?

Answer : Azure Virtual Machine Scale Sets (VMSS) allow you to create and manage a group of identical, load-balanced VMs that automatically increase or decrease in number based on demand or a schedule. VMSS ensures your application always has the right number of VMs – scaling out when demand rises and scaling in to reduce costs. All VMs in a scale set run the same OS image and configuration.

Q33 What is Azure App Service?

Answer : Azure App Service is a fully managed PaaS platform for building, deploying, and scaling web applications, REST APIs, and mobile backends. It supports multiple languages: .NET, Java, Node.js, Python, PHP, Ruby. Azure manages the underlying infrastructure (OS patching, capacity provisioning, load balancing). App Service Plan defines the region, number of VMs, and pricing tier.

Q34 What is Azure Container Instances (ACI)?

Answer : Azure Container Instances (ACI) is the fastest and simplest way to run containers in Azure without managing servers. It provides serverless containers – you submit a container image and Azure runs it. Ideal for simple applications, task automation, build jobs, and burst scenarios. ACI supports both Linux and Windows containers and can be connected to virtual networks.

Q35 What is Azure Kubernetes Service (AKS)?

Answer : Azure Kubernetes Service (AKS) is a managed Kubernetes orchestration service. Kubernetes automates deployment, scaling, and management of containerized applications. AKS handles the complex Kubernetes infrastructure (master node management, upgrades, monitoring) so you can focus on deploying and managing applications. AKS integrates with Azure AD, Container Registry, and Monitor.

Q36 What is Azure Functions?

Answer : Azure Functions is a serverless compute service that lets you run small pieces of code (functions) in the cloud without managing any infrastructure. Functions are triggered by events: HTTP requests, timer schedules, message queue messages, database changes, or blob uploads. You pay only when your code runs (consumption plan). Supports C#, Java, JavaScript, Python, PowerShell.

Q37 What is serverless computing?

Answer : Serverless computing is a cloud execution model where the cloud provider (Azure) automatically manages the infrastructure. You write code and Azure handles provisioning, scaling, and maintenance. 'Serverless' doesn't mean no servers — it means you don't see or manage them. Key characteristics: event-driven, automatic scaling, pay-per-execution, no idle cost. Azure Functions and Logic Apps are serverless services.

Q38 What is Azure Logic Apps?

Answer : Azure Logic Apps is a serverless service for automating workflows and integrating apps, data, and services. It uses a visual designer with pre-built connectors (200+) for services like Office 365, Salesforce, Twitter, SQL, and more. Logic Apps is no-code/low-code — business users can create automations without programming. It's ideal for enterprise integration, B2B workflows, and automating repetitive tasks.

Q39 What is Azure Virtual Desktop?

Answer : Azure Virtual Desktop (AVD) is a cloud-based desktop and app virtualization service. It delivers full Windows desktop experiences from Azure to any device. Features: multi-session Windows 10/11 (run multiple users on one VM), optimized Microsoft 365 experience, support for RemoteApp (stream individual apps), granular security controls, and automatic scaling. Ideal for remote work and BYOD scenarios.

Q40 What is an Azure Availability Set?

Answer : An Availability Set is a logical grouping of VMs that ensures they run across multiple fault domains and update domains within a single datacenter. Fault domains: separate racks with different power and network — protecting against hardware failure. Update domains: groups of VMs updated at different times during planned maintenance. Availability Sets provide 99.95% SLA compared to single VM SLA.

Q41 What is the difference between Availability Zones and Availability Sets?

Answer : Availability Sets distribute VMs across fault and update domains within a SINGLE datacenter — protects against hardware failures and planned maintenance. Availability Zones distribute VMs across SEPARATE physical datacenters within a region — protects against entire datacenter failures. Availability Zones provide higher SLA (99.99%) than Availability Sets (99.95%). Zones are preferred for most new deployments.

Q42 What are Azure VM pricing options?

Answer : Azure VMs have several pricing models: Pay-as-you-go: highest flexibility, no commitment, pay per second. Reserved Instances (1-3 years): up to 72% savings for predictable workloads. Spot VMs: use Azure's unused capacity at up to 90% discount — can be evicted with 30-second notice (ideal for batch jobs). Azure Hybrid Benefit: use existing Windows Server/SQL Server licenses to save up to 85%.

Q43 What is Azure Batch?

Answer : Azure Batch is a managed service for running large-scale parallel and high-performance computing (HPC) jobs. It automatically provisions VMs, installs applications, runs jobs as a queue, scales the pool as needed, and terminates VMs when done. Ideal for: rendering, financial risk modeling, genome analysis, scientific simulations. You pay only for the VMs used during job execution.

Q44 What is the Azure Marketplace?

Answer : The Azure Marketplace is an online store with thousands of pre-built application images, solutions, and services from Microsoft and third-party vendors. Categories include: VMs with pre-installed software, security solutions, developer tools, databases, AI/ML models, networking appliances, and SaaS applications. Marketplace solutions can be deployed to Azure with a few clicks, saving configuration time.

Q45 What is Azure Service Fabric?

Answer : Azure Service Fabric is a distributed systems platform for deploying and managing microservices and containers at scale. Unlike AKS (Kubernetes), Service Fabric includes built-in features for stateful services, actor models, and reliable collections. It's used by many Microsoft services (Azure SQL Database, Cosmos DB). Service Fabric can run on Azure, on-premises, or other clouds.

Q46 What is the difference between containers and VMs?

Answer : VMs virtualize hardware — each VM has its own OS kernel, taking GBs of storage and minutes to start. They provide strong isolation. Containers virtualize the OS — they share the host OS kernel, taking MBs of storage and starting in seconds. They're more lightweight but share the OS. VMs for: full OS control, diverse OSes. Containers for: microservices, portability, fast deployment.

Q47 What is Azure Container Registry (ACR)?

Answer : Azure Container Registry (ACR) is a managed Docker registry service for storing and managing container images and Helm charts. ACR integrates with AKS, ACI, and Azure Pipelines. It provides geo-replication to distribute images close to deployments, image scanning for vulnerabilities, and role-based access control. Use ACR to maintain a private registry of your organization's container images.

Q48 What is lift and shift migration?

Answer : Lift and shift (also called rehosting) is migrating applications from on-premises to the cloud with minimal or no changes to the application itself. You 'lift' the workload out of the current environment and 'shift' it to Azure VMs. It's the fastest migration approach — no refactoring needed. While not fully cloud-optimized, it provides immediate benefits of cloud (reduced hardware costs, scalability).

Q49 What is Azure Spring Apps?

Answer : Azure Spring Apps (formerly Azure Spring Cloud) is a fully managed service for running Spring Boot microservices applications. It handles the infrastructure so developers focus on application code. Features: automatic scaling, built-in service discovery, distributed tracing, blue-green deployments, and integration with Azure Monitor. Joint offering from Microsoft and VMware/Pivotal.

Q50 What is Azure VMware Solution?

Answer : Azure VMware Solution (AVS) lets organizations run VMware workloads natively on Azure dedicated bare-metal infrastructure. It's ideal for customers wanting to move VMware-based workloads to Azure without refactoring or retraining staff. AVS supports vSphere, vSAN, NSX-T, and vCenter environments. Microsoft and VMware jointly engineer, sell, and support the solution.

Q51 What is an Azure Resource Manager (ARM) template?

Answer : ARM templates are JSON files that define Azure infrastructure as code (IaC). They describe the resources to deploy (VMs, networks, databases) declaratively. ARM templates are idempotent — deploying the same template multiple times produces the same result. Benefits: consistent deployments, version control, automation, and reuse. Bicep is a newer domain-specific language that compiles to ARM JSON.

Q52 What is Azure Spot VM?

Answer : Azure Spot VMs use Microsoft's excess compute capacity at up to 90% discount compared to pay-as-you-go prices. The tradeoff: Azure can evict your Spot VM with 30 seconds notice when capacity is needed. Ideal for: batch processing, rendering, testing, stateless workloads that can be interrupted and resumed. Not suitable for business-critical workloads requiring guaranteed availability.

Q53 What is Azure Reserved VM Instances?

Answer : Azure Reserved VM Instances (RIs) are 1-year or 3-year commitments to use a specific VM type in a region, providing up to 72% savings compared to pay-as-you-go. RIs are a billing concept — they apply to running VMs that match the reservation attributes. You pay for the reservation whether or not you use it. Best for steady-state workloads with predictable usage patterns.

Q54 What is Azure Hybrid Benefit?

Answer : Azure Hybrid Benefit allows organizations to use existing on-premises Windows Server and SQL Server licenses (with active Software Assurance) on Azure, reducing cloud costs by up to 85%. For example, instead of paying for a new Windows Server license in Azure, you reuse your existing license. Combined with Reserved Instances, it offers maximum savings for license-included VM workloads.

Q55 What are Azure Compute Gallery (formerly Shared Image Gallery)?

Answer : Azure Compute Gallery is a service for storing, managing, and sharing VM images across your organization. It supports image versioning, replication to multiple regions, and role-based access sharing. Instead of recreating VMs from scratch, teams use standardized golden images with pre-installed software and configurations, ensuring consistency and reducing deployment time.

Q56 What is Azure dedicated hosts?

Answer : Azure Dedicated Hosts provide physical servers dedicated to a single Azure subscription. Unlike shared multi-tenant VMs, Dedicated Hosts give hardware isolation, visibility into physical server location, and control over maintenance windows. They support Azure Hybrid Benefit. Ideal for organizations with compliance requirements that mandate physical isolation from other customers.

Q57 What is proximity placement groups?

Answer : Proximity Placement Groups (PPGs) are logical groupings that ensure Azure resources (VMs, VMSS) are physically located close together within a datacenter. This minimizes network latency between VMs — important for latency-sensitive applications like high-performance databases, real-time processing, or tightly-coupled HPC workloads that require fast inter-node communication.

Q58 What is confidential computing in Azure?

Answer : Azure Confidential Computing protects data while it's being processed in memory (in use), in addition to at-rest and in-transit protection. It uses hardware-based Trusted Execution Environments (TEEs) called enclaves that encrypt data even from Azure operators. Use cases: multi-party computation, machine learning on sensitive data, blockchain, and processing regulated data (healthcare, financial).

Q59 What is Azure CycleCloud?

Answer : Azure CycleCloud is a tool for creating, managing, and optimizing HPC (High Performance Computing) clusters on Azure. It supports popular job schedulers (Slurm, PBS, LSF, Grid Engine) and dynamically scales compute nodes based on job queue depth. CycleCloud is ideal for research computing, financial modeling, oil and gas simulation, and other computationally intensive workloads.

Q60 What is the difference between Azure Functions Consumption Plan and Premium Plan?

Answer : Consumption Plan: automatic scaling, pay per execution, functions scale to zero when idle (cold start latency). Max execution time 10 minutes. Premium Plan: pre-warmed instances (no cold starts), VNET integration, extended execution time (60 minutes), more powerful instance sizes, support for always-on functions. Dedicated (App Service) Plan: functions run on your App Service VMs, suitable for long-running functions.

Section 3: Azure Core Services — Storage

Questions 61 – 90

Q61 What are the main Azure storage services?

Answer : Azure Storage offers four main data services: Azure Blob Storage (unstructured data — images, videos, backups, logs). Azure Files (fully managed file shares using SMB/NFS protocol). Azure Queue Storage (message queuing for decoupled applications). Azure Table Storage (NoSQL key-value store for structured data). All are accessed via a single Azure Storage Account and share the same scalable infrastructure.

Q62 What is Azure Blob Storage?

Answer : Azure Blob Storage is Microsoft's object storage solution for unstructured data — text, binary, images, videos, backups, log files, and data for analytics. 'Blob' stands for Binary Large Object. Three types of blobs: Block blobs (most common — files up to 190.7 TB). Append blobs (optimized for append operations like logging). Page blobs (random-access files used for Azure VM disks, up to 8 TB).

Q63 What are Azure Blob Storage access tiers?

Answer : Azure Blob Storage has four access tiers to optimize cost vs. access frequency: Hot tier: frequently accessed data, highest storage cost, lowest access cost. Cool tier: infrequently accessed (at least 30 days), lower storage cost. Cold tier: rarely accessed (at least 90 days). Archive tier: rarely accessed (180+ days), lowest storage cost, data is offline and takes hours to rehydrate. Tiers can be set per blob.

Q64 What is Azure Files?

Answer : Azure Files provides fully managed cloud file shares accessible via SMB (Server Message Block) and NFS (Network File System) protocols. Azure Files can be mounted by Windows, Linux, and macOS clients — both on-premises and in the cloud. Use cases: replacing on-premises file servers, 'lift and shift' of applications using file shares, shared application settings, and diagnostic logs accessible from multiple VMs.

Q65 What is Azure Queue Storage?

Answer : Azure Queue Storage is a service for storing large numbers of messages (up to 64 KB each, millions in a queue) to enable asynchronous communication between application components. One component writes messages to the queue; another reads and processes them independently. This decouples producers from consumers, improving reliability and scalability. Messages can be retained for up to 7 days.

Q66 What is Azure Table Storage?

Answer : Azure Table Storage is a NoSQL key-value store for structured, non-relational data. It stores data as rows (entities) with properties (columns), identified by a partition key and row key. Table Storage is fast and cost-effective for storing terabytes of structured data — device data, user data, metadata. It's now part of Azure Cosmos DB Table API, which offers better performance and global distribution.

Q67 What is a Storage Account?

Answer : An Azure Storage Account is the container that holds all Azure Storage data services (Blob, Files, Queue, Table). A storage account provides a unique namespace for your storage data (e.g., mystorageaccount.blob.core.windows.net). Types: Standard general-purpose v2 (most common), Premium (high-performance SSD-backed), BlockBlobStorage, and FileStorage. You choose redundancy level and access tier at the account level.

Q68 What are Azure Storage redundancy options?

Answer : Azure Storage offers four redundancy options: LRS (Locally Redundant Storage): 3 copies in one datacenter — cheapest, least protection. ZRS (Zone-Redundant Storage): 3 copies across 3 availability zones in one region. GRS (Geo-Redundant Storage): 6 copies — 3 in primary region (LRS) + 3 in secondary region. GZRS (Geo-Zone-Redundant Storage): ZRS in primary + 3 copies in secondary region — highest durability (16 nines).

Q69 What is Azure Disk Storage?

Answer : Azure Disk Storage provides persistent block storage volumes for Azure VMs, similar to physical hard drives. Disks persist independently of the VM lifetime. Types: Ultra Disk (highest performance — sub-millisecond latency, for databases), Premium SSD v2 (high performance, flexible), Premium SSD (high performance, production workloads), Standard SSD (cost-effective), Standard HDD (archival, low cost, infrequent access).

Q70 What is managed vs unmanaged disks?

Answer : Managed Disks: Azure manages storage account creation, sizing, and replication automatically. Easier to use, support Availability Zones and Sets, snapshot support, better security (encrypted by default). Recommended for all new deployments. Unmanaged Disks: customer manages the storage account where disk VHD files are stored. Legacy approach — Microsoft has deprecated unmanaged disks; new VMs should use managed disks.

Q71 What is Azure Blob Storage lifecycle management?

Answer : Lifecycle management policies automatically transition blobs between access tiers or delete them based on age or last access time. Example policy: move to Cool tier after 30 days, move to Cold tier after 90 days, move to Archive after 180 days, delete after 365 days. This helps automatically reduce storage costs for aging data without manual intervention.

Q72 What is Azure Data Lake Storage Gen2?

Answer : Azure Data Lake Storage Gen2 (ADLS Gen2) combines Azure Blob Storage with a hierarchical namespace for big data analytics. It's built on top of Blob Storage and adds: POSIX-compliant access control, hierarchical file system (directories), optimized driver for Hadoop/Spark/analytics. ADLS Gen2 is the recommended storage for big data analytics workloads using Azure Databricks, HDInsight, and Synapse Analytics.

Q73 What is Azure Storage Explorer?

Answer : Azure Storage Explorer is a free standalone application (Windows, Mac, Linux) for managing Azure storage resources visually. Features: browse and manage blobs, files, queues, and tables; upload/download data; manage access policies and SAS tokens; connect to storage accounts via connection string or Azure AD. Useful for data inspection, troubleshooting, and ad-hoc data management without writing code.

Q74 What is a Shared Access Signature (SAS)?

Answer : A Shared Access Signature (SAS) is a URI that grants restricted access to Azure Storage resources. It contains a token specifying: which resources are accessible, what permissions are granted (read/write/delete), and the validity time window. SAS tokens allow you to share access to storage without exposing your account keys. Types: Account SAS (full account access), Service SAS (specific service), User Delegation SAS (Azure AD-backed, most secure).

Q75 What is Azure Storage encryption?

Answer : Azure Storage automatically encrypts all data at rest using AES-256-bit encryption — this is always on and cannot be disabled. Encryption keys can be: Microsoft-managed keys (default, easiest), Customer-managed keys (CMK) stored in Azure Key Vault (greater control, auditable), or Customer-provided keys (you manage keys entirely). Data in transit is protected by HTTPS/TLS.

Q76 What is the difference between Blob Storage and Azure Files?

Answer : Blob Storage: object/unstructured storage accessed via REST API, Azure SDKs, or AzCopy. Best for large volumes of unstructured data, backups, streaming media, and serving static web content. Azure Files: file share accessed via SMB/NFS protocols, mountable as a network drive. Best for applications that use file system APIs, shared configurations, and lifting and shifting file-server workloads to the cloud.

Q77 What is AzCopy?

Answer : AzCopy is a command-line utility for copying data to, from, and between Azure Storage accounts. It supports Blob, Files, and Gen2. AzCopy is optimized for high throughput and can perform parallel data transfer. Use cases: large data migrations to Azure, copying data between storage accounts, synchronizing directories. It supports SAS tokens and Azure AD authentication.

Q78 What is Azure File Sync?

Answer : Azure File Sync is a service that centralizes file shares in Azure Files while keeping frequently accessed files cached on local Windows Server. It transforms Windows Server into a cache of your Azure file share. Features: multi-site sync (branch offices share the same files), cloud tiering (move infrequently accessed files to Azure to free local disk), disaster recovery (restore from Azure quickly).

Q79 What is immutable storage in Azure?

Answer : Immutable Blob Storage allows storing business-critical data in a WORM (Write Once, Read Many) state where data cannot be modified or deleted for a user-specified interval. Two policy types: Time-based retention (lock blobs for a defined period – e.g., 7 years for financial records). Legal hold (lock blobs indefinitely until the hold is cleared). Used for regulatory compliance (SEC 17a-4, HIPAA, FINRA).

Q80 What is Azure Backup?

Answer : Azure Backup is a cloud-based backup service that protects on-premises and Azure data. Supported sources: Azure VMs, SQL Server in VMs, Azure Files, SAP HANA databases, on-premises files via MARS agent. Azure Backup stores backups in Recovery Services Vaults with geo-redundant replication. Features: application-consistent backups, 14-day soft delete protection, role-based access, and long-term retention.

Q81 What is Azure Site Recovery?

Answer : Azure Site Recovery (ASR) is a disaster recovery service that replicates workloads from a primary site (on-premises or Azure region) to a secondary location. In the event of a disaster, you can failover to the secondary site with minimal data loss (near-zero RPO). ASR supports: Azure VM replication between regions, VMware/Hyper-V VM replication to Azure. It provides DR drills without impacting production.

Q82 What is the difference between LRS and GRS?

Answer : LRS (Locally Redundant Storage): stores 3 copies of data within a single datacenter in one region. Protects against disk and rack failures but NOT against datacenter-level disasters. Cheapest option. GRS (Geo-Redundant Storage): stores 3 copies in primary region (LRS) + asynchronously replicates to 3 copies in a paired secondary region. Provides durability across regional disasters. Data in secondary region is read-only unless failover.

Q83 What is Azure NetApp Files?

Answer : Azure NetApp Files is an enterprise-grade, high-performance, metered file storage service powered by NetApp hardware and software. It supports NFS and SMB protocols and provides ultra-low latency (sub-millisecond). Ideal for: SAP HANA, Oracle databases, high-performance computing, and latency-sensitive applications that require NFS/SMB with performance beyond what Azure Files provides.

Q84 What is object replication in Azure Blob Storage?

Answer : Object replication asynchronously copies block blobs between a source and destination storage account across regions. Use cases: minimize latency by serving reads from a geographically closer account, optimize compute workloads by processing data near its source, archive data to cold/archive tier in another region for cost savings. Requires versioning enabled on both accounts. Replication is asynchronous (eventual consistency).

Q85 What is blob versioning?

Answer : Blob versioning automatically maintains previous versions of a blob whenever it's modified or deleted. Each version has a unique version ID. If data is accidentally overwritten or deleted, you can restore a previous version. Versioning is integrated with lifecycle management — older versions can be automatically moved to cheaper tiers or deleted. Works alongside soft delete for comprehensive data protection.

Q86 What is soft delete in Azure Storage?

Answer : Soft delete protects blobs and containers from accidental deletion by retaining deleted data for a configurable period (1-365 days). Instead of permanent deletion, the data is marked as soft-deleted and hidden from normal operations. During the retention period, you can restore it. After the period expires, data is permanently deleted. Soft delete is free and can be enabled on any storage account.

Q87 What is Azure Data Box?

Answer : Azure Data Box is a physical device service for transferring large amounts of data (terabytes to petabytes) to Azure when network transfer is too slow or costly. Process: request a ruggedized device, copy data to it locally, ship it back to Microsoft, Microsoft uploads data to Azure, device is securely wiped. Variants: Data Box (up to 80 TB), Data Box Disk (up to 40 TB via SSD drives), Data Box Heavy (up to 1 PB).

Q88 What is the Azure Storage performance tiers?

Answer : Azure Storage has two performance tiers: Standard (HDD-backed): general-purpose storage, lower cost, supports all storage services (Blob, Files, Queue, Table). Suitable for most workloads. Premium (SSD-backed): high-performance, low-latency storage. Premium Block Blob (high transaction workloads), Premium File Shares (enterprise file shares requiring low latency), Premium Page Blobs (unmanaged VM disks requiring high I/O).

Q89 What is Azure CDN (Content Delivery Network)?

Answer : Azure CDN is a global network of servers that caches and delivers web content (static files, images, videos, scripts) close to users, reducing latency. When a user requests content, it's served from the nearest CDN edge server rather than the origin server. Azure CDN integrates with Azure Blob Storage, Web Apps, and Media Services. Providers: Microsoft, Verizon, and Akamai CDN profiles available.

Q90 What is geo-redundant storage failover?

Answer : GRS and GZRS normally keep secondary region data read-only. If the primary region becomes unavailable, you can initiate an account failover to make the secondary region the new primary. After failover, the account becomes LRS (the failed-over region has only one set of copies). You should re-enable geo-redundancy after recovery. Microsoft can also initiate failover if a region is completely unrecoverable.

Section 4: Azure Core Services — Networking

Questions 91 – 120

Q91 What is Azure Virtual Network (VNet)?

Answer : An Azure Virtual Network (VNet) is the fundamental building block of private networking in Azure. It enables Azure resources (VMs, databases, app services) to securely communicate with each other, the internet, and on-premises networks. A VNet is scoped to a region and subscription. You define its address space using CIDR notation (e.g., 10.0.0.0/16) and divide it into subnets.

Q92 What are subnets in Azure?

Answer : Subnets divide a VNet's address space into smaller segments. Resources in a subnet share the same network policies. Benefits: logical isolation (group resources by function — web tier, app tier, database tier), security (apply NSGs per subnet), and routing (custom routes per subnet). Azure reserves 5 IP addresses in each subnet (network address, broadcast, and 3 Azure internal addresses).

Q93 What is a Network Security Group (NSG)?

Answer : A Network Security Group (NSG) contains security rules that allow or deny inbound and outbound network traffic to Azure resources. Rules are based on: source/destination IP, port, protocol (TCP/UDP), and priority (lower number = higher priority). NSGs can be associated with subnets or individual VM network interfaces. Azure evaluates NSG rules from lowest priority number to highest until a match is found.

Q94 What is Azure VPN Gateway?

Answer : Azure VPN Gateway sends encrypted traffic between Azure VNet and on-premises networks over the public internet using IPsec/IKE protocols. Connection types: Site-to-Site (S2S): connects on-premises network to Azure VNet permanently. Point-to-Site (P2S): connects individual computers to Azure VNet (remote workers). VNet-to-VNet: connects Azure VNets in different regions. Bandwidth determined by gateway SKU.

Q95 What is Azure ExpressRoute?

Answer : Azure ExpressRoute provides private, dedicated connectivity between on-premises networks and Azure datacenters through a connectivity provider (not over the public internet). Benefits: up to 100 Gbps bandwidth, lower and more consistent latency than internet VPN, higher security (private circuit), SLA-backed reliability. More expensive than VPN but essential for high-bandwidth, mission-critical enterprise connections.

Q96 What is the difference between VPN Gateway and ExpressRoute?

Answer : VPN Gateway: encrypted connection over the public internet, lower cost (\$30-\$500/month), max 10 Gbps, higher latency. Good for moderate bandwidth and acceptable internet routing. ExpressRoute: private dedicated circuit bypassing the internet, higher cost (\$50-\$10,000/month), up to 100 Gbps, lower consistent latency, higher availability. Required for compliance scenarios where data cannot traverse the internet.

Q97 What is Azure Load Balancer?

Answer : Azure Load Balancer distributes inbound network traffic across multiple backend VMs to ensure high availability and scalability. It operates at Layer 4 (transport layer — TCP/UDP). Types: Public Load Balancer (distributes internet traffic to VMs). Internal Load Balancer (distributes traffic within a VNet). Load Balancer uses health probes to detect VM failures and automatically removes unhealthy instances from the rotation.

Q98 What is Azure Application Gateway?

Answer : Azure Application Gateway is a Layer 7 (HTTP/HTTPS) load balancer and web application firewall (WAF). Unlike Layer 4 Load Balancer, it can make routing decisions based on URL path, host headers, or cookies. Features: SSL/TLS termination (offloads SSL processing from backend VMs), WAF (protects against OWASP top 10 vulnerabilities, SQL injection, XSS), URL-based routing, session affinity, autoscaling.

Q99 What is Azure Traffic Manager?

Answer : Azure Traffic Manager is a DNS-based global traffic load balancer that distributes traffic across Azure regions or endpoints. It doesn't proxy traffic — it returns the IP of the best endpoint based on routing methods: Performance (closest endpoint), Priority (primary/failover), Weighted (distribute by percentage), Geographic (route by user location), Multivalue (return multiple healthy endpoints). Used for global high availability.

Q10
0 What is Azure Front Door?

Answer : Azure Front Door is a global Layer 7 load balancer and CDN with built-in WAF, SSL offloading, and URL-based routing. It uses Microsoft's global network to accelerate content delivery and provides automatic failover for globally distributed applications. Front Door can route users to the fastest backend, perform health checks, and protect applications with WAF policies. It replaces the older API of CDN+Traffic Manager for modern apps.

Q10
1 What is VNet Peering?

Answer : VNet Peering connects two Azure VNets so resources in each VNet can communicate with each other using private IP addresses, as if they're on the same network. Traffic between peered VNets travels over Microsoft's backbone network (not internet) – low latency and high bandwidth. Local peering: within the same region. Global peering: across different regions. Peering is non-transitive – VNet A peers B and C, but B and C can't communicate unless directly peered.

**Q10
2** What is Azure DNS?

Answer : Azure DNS is a hosting service for DNS domains using Azure infrastructure. It provides name resolution using Microsoft's global network of DNS servers. You can manage DNS records (A, AAAA, CNAME, MX, TXT, SRV) for your custom domains using Azure Portal, PowerShell, or CLI. Azure DNS is integrated with Azure services and supports both public DNS zones and private DNS zones for internal name resolution within VNets.

**Q10
3** What is a Private DNS Zone?

Answer : Azure Private DNS Zone provides name resolution for resources within Azure VNets without requiring a custom DNS server. Private zones are only accessible within linked VNets. Use cases: naming VMs with friendly names instead of IP addresses, resolving Azure PaaS service names privately (e.g., mydb.privatelink.database.windows.net). Private DNS zones support automatic registration of VM names when linked to a VNet.

**Q10
4** What is Azure Firewall?

Answer : Azure Firewall is a managed, cloud-native network security service that protects Azure VNet resources. It's a stateful firewall with built-in high availability and unrestricted scalability. Features: FQDN (domain name) filtering, network filtering rules, threat intelligence integration (block known malicious IPs), TLS inspection, IDPS (Intrusion Detection and Prevention), centralized management via Azure Firewall Manager.

**Q10
5** What is Azure DDoS Protection?

Answer : Azure DDoS Protection defends Azure resources from Distributed Denial of Service attacks. Two tiers: Basic (free, always-on, automatically protects all Azure services from common network-layer attacks). Standard (premium, adaptive tuning based on your traffic patterns, advanced attack analytics, cost protection guarantee, 24/7 DDoS rapid response team, covers all resources in protected VNet). Standard is recommended for critical applications.

**Q10
6** What is Azure Bastion?

Answer : Azure Bastion is a fully managed PaaS service providing secure RDP and SSH connectivity to VMs directly from the Azure Portal over TLS (HTTPS). VMs don't need public IP addresses — Bastion connects through the browser. This eliminates exposure of VMs to the internet for management access, protecting against port scanning and brute force attacks. Bastion is deployed inside a VNet's AzureBastionSubnet.

Q107 What is a Network Virtual Appliance (NVA)?

Answer : A Network Virtual Appliance (NVA) is a VM running network functions such as firewalls, routers, WAN optimizers, or load balancers from third-party vendors (Check Point, Palo Alto, Fortinet, Cisco) deployed from Azure Marketplace. NVAs extend Azure's native networking capabilities with specialized features not available in Azure-native services. They run as VMs in your VNet and require UDRs to route traffic through them.

Q108 What is a User Defined Route (UDR)?

Answer : User Defined Routes (UDRs) allow you to override Azure's default system routes to control how traffic flows within a VNet. Example: force all internet-bound traffic through an NVA firewall instead of routing directly to the internet. You create a Route Table with custom routes and associate it with a subnet. Common use: hub-and-spoke network topology where all inter-spoke traffic passes through a central hub firewall.

Q109 What is Azure Virtual WAN?

Answer : Azure Virtual WAN is a networking service that provides optimized and automated branch connectivity to Azure. It simplifies connectivity for large enterprises with many branch offices by providing a unified hub connecting branches, VNets, ExpressRoute, and VPN in a single managed service. Features: any-to-any connectivity, built-in firewall integration, optimized routing, and simplified management of global networking at scale.

Q110 What is a Service Endpoint?

Answer : A VNet Service Endpoint extends your VNet's private address space and identity to Azure services (Storage, SQL, Key Vault, etc.) over Microsoft's backbone network. With service endpoints, traffic from your VNet to the Azure service stays on the Microsoft network (not the internet). You can restrict service access to your VNet only, improving security. Service endpoints are subnet-level settings.

Q111 What is Private Link / Private Endpoint?

Answer : Azure Private Link enables access to Azure PaaS services (Storage, SQL, Cosmos DB, Key Vault) and partner services over a private endpoint in your VNet. A Private Endpoint assigns a private IP from your VNet to the Azure service, making it appear as if the service is inside your VNet. Traffic never leaves the Microsoft backbone. More secure than service endpoints because it's specific to a resource instance.

**Q11
2** What is the difference between Service Endpoint and Private Endpoint?

Answer : Service Endpoint: enables traffic from subnet to Azure service over Microsoft backbone, but the service still has a public IP and can be accessed from internet (you restrict access via firewall rules). Private Endpoint: assigns a private IP in your VNet to a specific service instance — service appears to have no public IP to your traffic. Private endpoints provide per-resource private access; service endpoints are subnet-wide.

**Q11
3** What is Azure NAT Gateway?

Answer : Azure NAT (Network Address Translation) Gateway provides outbound internet connectivity for resources in a subnet. Instead of each VM having its own public IP, all outbound connections share the NAT Gateway's public IP(s) with up to 64,000 connections per IP. Benefits: simplified outbound IP management, consistent predictable public IPs, no need for individual VM public IPs, high throughput and low latency outbound connectivity.

**Q11
4** What is Azure Network Watcher?

Answer : Azure Network Watcher provides tools for monitoring, diagnosing, and gaining insights into Azure network resources. Key features: Connection Monitor (continuous connectivity monitoring), IP Flow Verify (check if traffic is allowed/denied by NSG), Next Hop (view routing decisions), VPN Diagnostics (troubleshoot VPN gateways), Packet Capture (capture packets from VMs), NSG Flow Logs (log traffic through NSGs for analysis).

**Q11
5** What are Availability Zones for networking?

Answer : Availability Zone-redundant networking services deploy across multiple zones automatically: Zone-redundant Standard Load Balancer and Standard Public IP distribute traffic across zones. Azure Firewall with zone redundancy. Azure VPN Gateway zone-redundant SKUs. Zone-redundant networking ensures the service continues to operate even if an entire datacenter zone fails, providing higher SLA than single-zone deployments.

**Q11
6** What is the hub-and-spoke network topology?

Answer : Hub-and-spoke is a common Azure network architecture where a central hub VNet connects to multiple spoke VNets via peering. The hub contains shared services: firewall/NVA, VPN/ExpressRoute gateway, Active Directory, monitoring. Spokes contain workloads. Traffic between spokes passes through the hub (using UDRs). Benefits: centralized security inspection, shared connectivity, reduced management overhead, and cost savings.

Q117 What is Azure CDN compared to Azure Front Door?

Answer : Azure CDN: global caching network for static content (images, CSS, JS), reduces origin server load and delivers content from edge locations near users. Works at Layer 7 but primarily for caching. Azure Front Door: global HTTP load balancer with CDN caching, WAF, health probes, and intelligent routing. Front Door adds application acceleration, routing rules, and protection to CDN capabilities. Front Door is the modern recommended service.

Q118 What is IP addressing in Azure?

Answer : Azure resources can have public or private IPs. Private IPs: assigned from VNet address space, used for internal communication. Public IPs: globally routable internet addresses assigned to resources. Public IP SKUs: Basic (open by default, no zone redundancy, being retired), Standard (secure by default, zone redundant, required for Zone redundant resources). Azure dynamically assigns IPs by default; static IPs are also available.

Q119 What is peering vs VPN for VNet connectivity?

Answer : VNet Peering: direct Microsoft backbone connection between VNets, low latency, high bandwidth, no gateway required, no encryption overhead. Best for: Azure-to-Azure VNet connectivity. VPN Gateway VNet-to-VNet: encrypted tunnel between VNets, requires gateway in each VNet, slightly higher latency. Required when: VNets are in different Azure Active Directory tenants, or when you need encrypted traffic between VNets (compliance).

Q120 What is Azure Private DNS Resolver?

Answer : Azure DNS Private Resolver is a cloud-native, fully managed DNS resolution service that allows on-premises networks to query Azure Private DNS zones without deploying custom DNS servers (like Windows Server DNS). It provides inbound endpoints (on-premises queries DNS in Azure) and outbound endpoints (Azure resources forward DNS queries to on-premises DNS). Eliminates the need to manage custom DNS server VMs for hybrid DNS.

Section 5: Azure Identity, Access & Security

Questions 121 – 150

Q12₁ What is Microsoft Entra ID (formerly Azure Active Directory)?

Answer : Microsoft Entra ID (formerly Azure AD) is Microsoft's cloud-based identity and access management service. It provides: authentication (verify who you are), authorization (what you can access), single sign-on (SSO) for thousands of cloud apps, multi-factor authentication (MFA), conditional access, and B2B/B2C collaboration. It's the backbone of security for Microsoft 365, Azure, and thousands of SaaS applications.

Q12₂ What is the difference between Azure AD and on-premises Active Directory?

Answer : On-premises Active Directory (AD DS): manages identities in a corporate network using LDAP, Kerberos, and NTLM. Designed for domain-joined Windows PCs. Azure AD (Entra ID): cloud-based, uses HTTP/HTTPS, OAuth 2.0, OpenID Connect, SAML. Manages access to cloud apps. No direct domain-join for PCs. Azure AD Connect synchronizes on-premises AD identities to Azure AD for hybrid scenarios.

Q12₃ What is Azure AD Connect?

Answer : Azure AD Connect is a tool that synchronizes on-premises Active Directory identities to Azure AD, enabling hybrid identity. Features: Password Hash Synchronization (PHS) — syncs password hashes to Azure AD, users log in with same credentials. Pass-Through Authentication (PTA) — validates passwords against on-premises AD in real time. Federation with AD FS — delegates authentication to on-premises AD FS. Choose based on security and compliance needs.

Q12₄ What is Multi-Factor Authentication (MFA)?

Answer : MFA requires users to provide two or more verification factors to access a resource — something you know (password), something you have (phone/authenticator app), something you are (fingerprint). Azure AD MFA protects against 99.9% of account compromise attacks. MFA can be enforced via Conditional Access policies, per-user settings, or Security Defaults (free MFA for all admins). Methods: Authenticator app, SMS, phone call, FIDO2 keys.

Q12₅ What is Conditional Access?

Answer : Conditional Access is Azure AD's policy engine that controls access to apps and resources based on conditions and signals. Signals include: user/group, device compliance, location (IP/country), application being accessed, and real-time risk detection. Access decisions: Allow, Block, or Require additional verification (MFA, compliant device). Example policy: 'Require MFA for all users accessing Azure Portal from outside the corporate network.'

Q12
6 What is Azure Role-Based Access Control (RBAC)?

Answer : Azure RBAC controls who has access to Azure resources, what they can do, and what resources they can access. It uses role assignments: Security Principal (who: user, group, service principal, managed identity) + Role Definition (what: built-in or custom roles with allowed actions) + Scope (where: management group, subscription, resource group, or resource). RBAC is additive – multiple role assignments combine permissions.

Q12
7 What are the built-in Azure RBAC roles?

Answer : Key built-in roles: Owner – full access including ability to assign roles. Contributor – full access except cannot assign roles or manage access. Reader – view all resources but cannot make changes. User Access Administrator – manage user access to Azure resources (not resource management). Resource-specific roles: Virtual Machine Contributor, Storage Blob Data Contributor, Key Vault Reader, etc. Prefer least-privilege built-in roles over Owner.

Q12
8 What is the principle of least privilege?

Answer : The principle of least privilege means granting users, applications, and services only the minimum permissions necessary to perform their required tasks. In Azure, this means: using Reader instead of Contributor when modification isn't needed, scoping role assignments to specific resources rather than subscriptions, using managed identities instead of credentials, and regularly reviewing and removing unused access.

Q12
9 What is Zero Trust security model?

Answer : Zero Trust is a security model based on the principle 'never trust, always verify.' Instead of assuming everything inside a corporate network is safe, Zero Trust verifies every access request as if it originates from an open network. Principles: Verify explicitly (always authenticate and authorize based on all available data points), Use least privilege access, Assume breach (minimize blast radius, segment access). Azure implements Zero Trust through Entra ID, Defender, and Purview.

Q13
0 What is Azure Key Vault?

Answer : Azure Key Vault is a cloud service for securely storing and managing: Secrets (API keys, passwords, connection strings), Cryptographic keys (used for data encryption), and SSL/TLS Certificates. Key Vault provides: centralized secret management (apps access secrets via API, not hardcoded), access control via Azure AD and RBAC, audit logging of all access, automatic key rotation, and HSM (Hardware Security Module) backing for premium tier.

**Q13
1** What is a Managed Identity?

Answer : A Managed Identity is an automatically managed identity in Azure AD that applications can use to authenticate to Azure services – without storing credentials in code. Types: System-assigned managed identity (tied to a specific Azure resource's lifecycle, deleted when resource is deleted). User-assigned managed identity (standalone identity, can be assigned to multiple resources). Use managed identities instead of service principals with secrets when possible.

**Q13
2** What is Azure Defender (Microsoft Defender for Cloud)?

Answer : Microsoft Defender for Cloud (formerly Azure Security Center + Azure Defender) is a unified security management and threat protection service. Features: Secure Score (measures security posture), Security recommendations (actionable steps to improve security), threat detection for VMs, SQL, Kubernetes, storage, DNS, and Key Vault, regulatory compliance dashboard, and just-in-time VM access to reduce attack surface.

**Q13
3** What is Microsoft Sentinel?

Answer : Microsoft Sentinel is a cloud-native SIEM (Security Information and Event Management) and SOAR (Security Orchestration, Automation, and Response) solution. It collects data from all sources (users, devices, apps, infrastructure), uses AI to detect threats, investigates incidents with built-in threat intelligence, and automates responses. Sentinel eliminates the need to manage on-premises SIEM infrastructure.

**Q13
4** What is Azure Policy?

Answer : Azure Policy is a governance service that enforces organizational standards and assesses compliance across Azure resources. Policies define rules (e.g., 'all storage accounts must use HTTPS', 'VMs must be in approved regions'). Policy effects: Deny (block non-compliant resources), Audit (log non-compliant resources), Append (add fields to resources), DeployIfNotExists (deploy remediation resources). Policies can be grouped into Initiatives.

**Q13
5** What is the difference between Azure Policy and RBAC?

Answer : RBAC controls WHO can perform actions on resources — it's about user permissions and access. Azure Policy controls WHAT resources can be created and their properties — it's about resource compliance. Example: RBAC lets a user create VMs, Azure Policy ensures VMs are only created in allowed regions and with specific sizes. They work together: RBAC allows actions, Policy governs the resulting resource properties.

**Q13
6** What is Microsoft Purview?

Answer : Microsoft Purview (formerly Azure Purview) is a unified data governance service for managing and governing data across cloud and on-premises environments. Features: Data Map (catalog and classify all data assets), Data Catalog (make data discoverable), Data Lineage (track data from source to destination), Information Protection (label and protect sensitive data), Data Loss Prevention (DLP), and compliance management across Microsoft 365 and Azure.

**Q13
7** What is Azure Blueprints?

Answer : Azure Blueprints enables cloud architects to define a repeatable set of Azure resources that implements and adheres to an organization's standards, patterns, and requirements. A blueprint bundles: ARM templates, Policy assignments, RBAC role assignments, and resource groups. Blueprints are tracked — you can see which resources were deployed from a blueprint and update them together. Useful for deploying compliant environments at scale.

**Q13
8** What is Microsoft Defender for Cloud Secure Score?

Answer : Secure Score in Microsoft Defender for Cloud is a numerical measurement of an organization's security posture (0-100%). It aggregates security recommendation findings into a score. Completing security recommendations increases the score. Each recommendation has a point value based on severity. Secure Score helps prioritize security improvements and track progress over time. Higher score = better security posture.

**Q13
9** What is Azure Active Directory B2B?

Answer : Azure AD B2B (Business-to-Business) collaboration allows organizations to invite external users (partners, vendors, consultants) to access Azure resources and Microsoft 365 apps using their existing identities (Microsoft, Google, social, or organizational accounts). External users are represented as guest users in the directory. No need to manage external credentials — guests authenticate through their own identity provider.

**Q14
0** What is Azure AD B2C?

Answer : Azure AD B2C (Business-to-Consumer) is a customer identity and access management (CIAM) solution for consumer-facing applications. It allows customers to sign up, sign in, and manage their profiles using social identities (Google, Facebook, Apple, Amazon) or local accounts. B2C is highly customizable with branded sign-in pages and custom policies. Separate from Azure AD — designed for millions of external consumer identities.

**Q14
1** What is Privileged Identity Management (PIM)?

Answer : Azure AD Privileged Identity Management (PIM) manages, controls, and monitors access to important resources. Key features: Just-in-time privileged access (users request elevated access for a limited time period), Access reviews (periodically review who has privileged access), Audit history of privileged access, Require approval and MFA for role activation. PIM reduces the risk of over-privileged accounts being compromised.

**Q14
2** What is Identity Protection?

Answer : Azure AD Identity Protection uses machine learning to detect and respond to suspicious sign-in activities. It detects: impossible travel, leaked credentials (known password breaches), sign-ins from anonymous IPs, and unfamiliar locations. When risk is detected, Identity Protection can: block access, require MFA, require password reset, or alert admins. Requires Azure AD Premium P2 license.

**Q14
3** What are security defaults in Azure AD?

Answer : Security Defaults are a set of pre-configured security settings that Microsoft recommends for all organizations — free with any Azure AD license. They enable: MFA for all users and admins, blocking legacy authentication protocols (which can't use MFA), requiring MFA for privileged admin operations. Security Defaults are disabled when Conditional Access is used (as CA provides more granular control).

**Q14
4** What is the difference between authentication and authorization?

Answer : Authentication (AuthN): verifying the identity of a user or service — 'Who are you?' Uses credentials (password, certificate, MFA). In Azure: handled by Azure AD. Authorization (AuthZ): determining what an authenticated identity can do — 'What are you allowed to do?' Uses permissions and roles. In Azure: handled by RBAC for Azure resources and app-specific permissions for applications. Authentication always precedes authorization.

**Q14
5** What is FIDO2 / Passwordless authentication?

Answer : FIDO2 is an open authentication standard supporting passwordless and phishing-resistant authentication. Azure AD supports: Windows Hello for Business (biometric/PIN), FIDO2 security keys (YubiKey, etc.), and Microsoft Authenticator app (phone sign-in). Passwordless removes the weakest security element (passwords) while improving user experience. FIDO2 keys are the most phishing-resistant option as they're bound to specific websites.

Q14
6 What is Azure AD Single Sign-On (SSO)?

Answer : SSO allows users to authenticate once and access multiple applications without re-entering credentials. Azure AD provides SSO for: Microsoft 365, thousands of pre-integrated SaaS apps (Salesforce, ServiceNow, Workday), on-premises apps via Application Proxy, and custom apps. SSO protocols: SAML 2.0, OpenID Connect/OAuth 2.0, Header-based, Integrated Windows Authentication. SSO improves productivity and security (fewer passwords = fewer breaches).

Q14
7 What is Just-In-Time (JIT) VM Access?

Answer : Just-In-Time VM access (part of Microsoft Defender for Cloud) locks down inbound traffic to Azure VMs by default. When admins need to connect (RDP/SSH), they request access for a specific time period (1-3 hours). JIT dynamically creates an NSG rule allowing the connection from only the requester's IP. After the time expires, the rule is removed. This dramatically reduces exposure to port scanning and brute-force attacks.

Q14
8 What is Microsoft Defender for Identity?

Answer : Microsoft Defender for Identity (formerly Azure ATP) is a cloud-based security solution that monitors on-premises Active Directory signals to detect and investigate advanced threats, compromised identities, and malicious insider actions. It detects: Pass-the-Hash, Pass-the-Ticket, Kerberoasting, lateral movement, and reconnaissance attacks. It integrates with Microsoft 365 Defender for unified security investigation.

Q14
9 What is Azure Information Protection (AIP)?

Answer : Azure Information Protection (now part of Microsoft Purview Information Protection) classifies and protects documents and emails by applying labels. Labels can automatically encrypt documents, add watermarks, and restrict forwarding. Sensitivity labels persist with the document regardless of location. AIP supports built-in and custom classification rules. It helps prevent accidental data leakage of confidential documents.

Q15
0 What is Microsoft Entra Permissions Management?

Answer : Microsoft Entra Permissions Management (formerly CloudKnox) is a Cloud Infrastructure Entitlement Management (CIEM) solution providing visibility into permissions across multi-cloud environments (Azure, AWS, Google Cloud). It identifies over-privileged identities, detects anomalous permission usage, and right-sizes permissions based on actual usage patterns. Helps achieve least-privilege access at cloud scale.

Section 6: Azure Management & Governance

Questions 151 – 180

**Q15
1** What is the Azure management hierarchy?

Answer : Azure has four levels of management hierarchy: Management Groups (top level — organize subscriptions, apply governance across multiple subs), Subscriptions (billing unit and policy boundary, contains resource groups), Resource Groups (logical containers for related resources), Resources (individual services like VMs, databases). Policies and RBAC applied at higher levels are inherited by all lower levels.

**Q15
2** What is an Azure Subscription?

Answer : An Azure subscription is a logical container for provisioning Azure resources and a billing boundary. All resources in a subscription are billed together to one account. Organizations can have multiple subscriptions to: separate billing by department, apply different policies per environment (dev/prod), isolate resources for security. Subscriptions have resource limits (quotas) that can be increased by contacting Microsoft Support.

**Q15
3** What are Management Groups?

Answer : Management Groups are containers for organizing Azure subscriptions into a hierarchy for unified governance and compliance. You can apply Azure Policies and RBAC role assignments to management groups — all subscriptions and resources within inherit these settings. The hierarchy can have up to 6 levels of nesting (below the root management group). Every Azure AD tenant has one root management group.

**Q15
4** What is a Resource Group?

Answer : A Resource Group is a logical container that holds related Azure resources. All resources in a group share the same lifecycle — you can deploy, update, and delete them together. Resources must be in exactly one resource group, one region. Resource groups can contain resources from multiple regions. RBAC and Azure Policy can be scoped to resource groups. Use resource groups to organize resources by application, environment, or team.

**Q15
5** What is Azure Resource Manager (ARM)?

Answer : Azure Resource Manager (ARM) is the deployment and management service for Azure. All operations (create, update, delete resources) go through ARM whether using Portal, CLI, PowerShell, or APIs. ARM provides: consistent management layer, resource grouping, template-based deployment, RBAC integration, dependency management (deploy resources in correct order), and tagging for organization and billing.

**Q15
6** What are Azure Tags?

Answer : Azure Tags are name-value pairs (metadata) attached to Azure resources for organization and management. Examples: Environment:Production, Department:Finance, CostCenter:12345. Uses: cost tracking and allocation (filter billing by tag), automation (target resources with specific tags), compliance (identify untagged resources with Policy), and governance (enforce tagging standards). Resources can have up to 50 tags.

**Q15
7** What is the Azure Portal?

Answer : The Azure Portal is a web-based graphical interface at portal.azure.com for managing Azure resources. Features: resource creation and management, monitoring dashboards, cost analysis, support requests, and personalized dashboards. The Portal provides access to all Azure services and integrates with Azure AD for authentication. It's suitable for individual resource management and exploration but less efficient than CLI/PowerShell for automation.

**Q15
8** What is Azure Cloud Shell?

Answer : Azure Cloud Shell is an interactive, browser-based shell for managing Azure resources — accessible from the Azure Portal, Azure mobile app, or shell.azure.com. It provides: pre-authenticated Bash or PowerShell environment, pre-installed Azure CLI, Azure PowerShell, Terraform, Ansible, kubectl, and other tools. Files persist in Azure Files storage. Eliminates need to install tools locally.

**Q15
9** What is Azure CLI?

Answer : Azure CLI (Command-Line Interface) is a cross-platform command-line tool for managing Azure resources. Available on Windows, macOS, and Linux. Commands start with 'az': 'az vm create', 'az group list', 'az storage account show'. Benefits: scriptable for automation, faster than Portal for repetitive tasks, consistent across platforms, version-controllable scripts. Authenticate with 'az login'. Available in Azure Cloud Shell.

**Q16
0** What is Azure PowerShell?

Answer : Azure PowerShell is a module with cmdlets for managing Azure resources from PowerShell (Windows, macOS, Linux). Commands use Azure-specific verbs: 'New-AzVM', 'Get-AzResourceGroup', 'Remove-AzStorageAccount'. Preferred by Windows administrators and .NET developers. Benefits: full PowerShell scripting capabilities, pipeline support, object-oriented output. Available in Azure Cloud Shell alongside Azure CLI.

Q16 1 What is Azure Cost Management?

Answer : Azure Cost Management + Billing is a set of tools for monitoring, analyzing, and optimizing Azure spending. Features: Cost analysis (visualize spending by service, region, or tag), Budgets (set spending thresholds with alert notifications), Cost alerts (notify when spending exceeds thresholds), Recommendations (identify idle/underutilized resources), Azure Advisor cost recommendations, and invoicing management.

Q16 2 What is the Azure Pricing Calculator?

Answer : The Azure Pricing Calculator (azure.microsoft.com/pricing/calculator) estimates costs for Azure services before deployment. You add services, configure their settings (region, tier, quantity, hours), and get a monthly cost estimate. It helps compare pricing for different configurations and regions. The calculator doesn't access your actual Azure usage — it's purely for planning and estimation purposes.

Q16 3 What is the Azure TCO Calculator?

Answer : The Azure Total Cost of Ownership (TCO) Calculator estimates how much you can save by migrating on-premises workloads to Azure. You enter on-premises infrastructure details (servers, storage, networking, electricity, labor costs) and Azure calculates the equivalent Azure costs plus savings over 3 or 5 years. Typical savings: 30-90%. Useful for building a business case for cloud migration.

Q16 4 What is Azure Advisor?

Answer : Azure Advisor is a personalized cloud consultant that analyzes your Azure usage and configurations, providing actionable recommendations across 5 categories: Reliability (improve availability), Security (fix vulnerabilities — integrates with Defender for Cloud), Performance (improve speed), Cost (reduce spending — identify idle resources, rightsize VMs), Operational Excellence (improve deployment and management processes).

Q16 5 What is Azure Monitor?

Answer : Azure Monitor is a comprehensive monitoring service for collecting, analyzing, and acting on telemetry from Azure and on-premises environments. Components: Metrics (numerical performance data), Logs (text-based event data in Log Analytics workspace), Alerts (notify on conditions), Application Insights (APM for web applications), Dashboards (visualize metrics), Autoscale (scale resources based on metrics).

Q16₆ What is Azure Log Analytics?

Answer : Azure Log Analytics is a workspace service within Azure Monitor for querying and analyzing log data. It uses Kusto Query Language (KQL) to query large datasets efficiently. Sources: Azure Monitor Logs, Microsoft Sentinel, Defender for Cloud, Azure Activity Log, VM agents, and custom sources. Log Analytics enables: complex log queries, dashboards, alerts, and correlation across multiple data sources.

Q16₇ What is Application Insights?

Answer : Application Insights is an Application Performance Management (APM) service within Azure Monitor for live web applications. It monitors: request rates and response times, failure rates and exceptions, page views and load performance, user and session counts, custom events, and dependency tracking (external calls). Supports .NET, Java, Node.js, Python applications. Helps detect and diagnose performance anomalies automatically.

Q16₈ What is the Azure Service Health?

Answer : Azure Service Health is a personalized dashboard providing information about Azure service health in the regions you use. Three components: Azure Status (global Azure outages), Service Health (personalized view of outages and maintenance affecting your subscriptions and regions), Resource Health (health of your specific resources – is your VM accessible?). Supports health alerts to notify via email, SMS, or webhook.

Q16₉ What is Azure Service Trust Portal?

Answer : The Service Trust Portal (servicetrust.microsoft.com) is where Microsoft publishes information about compliance, security, and privacy practices for Microsoft cloud services. It provides: audit reports (SOC 1/2/3, ISO 27001, PCI DSS), compliance guides, whitepapers, penetration test results, and the Microsoft Privacy Statement. Useful for customers needing evidence of Microsoft's compliance certifications.

Q17₀ What is Microsoft Privacy Statement?

Answer : The Microsoft Privacy Statement explains what personal data Microsoft collects, how it's used and shared, and your rights regarding that data. For Azure specifically: Microsoft commits to GDPR compliance, data is used only to provide services (not for advertising), customers own their data, Microsoft is a data processor (customers are data controllers), data can be exported or deleted, and Microsoft provides transparency reports.

Q171 What is the Azure compliance documentation?

Answer : Azure compliance documentation (docs.microsoft.com/azure/compliance) provides information on Azure's compliance with regulatory frameworks and industry standards. Azure has the largest compliance certification portfolio: 100+ certifications including ISO 27001, SOC 1/2/3, PCI DSS Level 1, HIPAA, FedRAMP, GDPR, and country-specific regulations. Compliance Manager helps organizations track their own compliance against these frameworks.

Q172 What is Azure Migrate?

Answer : Azure Migrate is a centralized hub for discovering, assessing, and migrating on-premises workloads to Azure. Key tools: Discovery and Assessment (discover VMware/Hyper-V/physical servers, assess Azure readiness, estimate costs), Server Migration (replicate and migrate VMs), Database Migration Service (migrate SQL Server, MySQL, PostgreSQL to Azure), App Service Migration Assistant (migrate web apps). Free tool — pay only for Azure resources used.

Q173 What is Azure Arc?

Answer : Azure Arc extends Azure management capabilities to any infrastructure — on-premises, multi-cloud, and edge. It allows you to manage servers (Windows/Linux), Kubernetes clusters, SQL Server, and Azure data services running outside Azure using the same Azure tools, governance (Policy, RBAC), and security. Azure Arc brings Azure's control plane to hybrid and multi-cloud environments.

Q174 What is the Azure mobile app?

Answer : The Azure mobile app (iOS and Android) lets you manage Azure resources from your smartphone. Features: monitor resource health and metrics, view alerts, start/stop/restart VMs, browse resource properties, run Cloud Shell (CLI/PowerShell), receive push notifications for alerts. Useful for on-call scenarios where you need to quickly check or respond to Azure resource issues without a laptop.

Q175 What are Azure Locks?

Answer : Azure Locks prevent accidental deletion or modification of critical resources. Two lock types: Read-only lock (CanNotDelete is wrong – read-only prevents modifications AND deletions). Delete lock (CanNotDelete) – prevents deletion but allows modifications. Locks are inherited by child resources. Locks can be set at subscription, resource group, or resource level. Owner role is required to create/delete locks.

**Q17
6** What is Azure DevTest Labs?

Answer : Azure DevTest Labs is a service for quickly creating development and test environments, minimizing waste and controlling costs. Features: self-service VM provisioning from pre-configured templates, automatic VM shutdown schedules, spending limits and quotas per user, reusable environments (formulas and artifacts), and integration with CI/CD pipelines. Developers get VMs in minutes without waiting for IT, with guardrails to prevent overspending.

**Q17
7** What is Bicep?

Answer : Bicep is a domain-specific language (DSL) for deploying Azure resources declaratively. It's a transparent abstraction over ARM JSON templates – all Bicep code compiles to ARM JSON. Benefits: cleaner syntax than ARM JSON, no state management, first-class VS Code support with IntelliSense, modular code with modules, type safety. Microsoft recommends Bicep as the preferred IaC language for Azure-only deployments.

**Q17
8** What is Terraform for Azure?

Answer : Terraform is a popular open-source IaC tool from HashiCorp that supports Azure through the AzureRM provider. Unlike Bicep (Azure-only), Terraform is cloud-agnostic – same tool for Azure, AWS, and Google Cloud. Terraform maintains a state file tracking deployed resources. Use Terraform when: managing multi-cloud environments, existing Terraform expertise in team, or needing third-party provider ecosystem. Azure provides native Terraform support.

**Q17
9** What is the Azure Well-Architected Framework?

Answer : The Azure Well-Architected Framework is a set of guiding tenets for building high-quality Azure solutions across five pillars: Reliability (withstand failures), Security (protect data and systems), Cost Optimization (maximize value), Operational Excellence (deploy efficiently), Performance Efficiency (scale efficiently). It provides best practices, architecture guidance, and review tools to help design and improve Azure workloads.

**Q18
0** What is the Cloud Adoption Framework (CAF)?

Answer : The Microsoft Cloud Adoption Framework for Azure is a collection of documentation, technical guidance, best practices, and tools that help organizations in their cloud journey. Phases: Strategy (define motivations), Plan (align teams and technologies), Ready (prepare environment), Adopt (migrate and innovate), Govern (manage risk), Manage (operations). CAF guides organizations from initial cloud planning through mature cloud operations.

Section 7: Azure AI, ML & Modern Services

Questions 181 – 210

Q18₁ What is Azure AI Services (Cognitive Services)?

Answer : Azure AI Services (formerly Cognitive Services) is a collection of pre-built AI APIs and services that add intelligent capabilities to applications without ML expertise. Categories: Vision (Computer Vision, Face, Custom Vision), Language (Text Analytics, Translator, LUIS, Language Understanding), Speech (Speech-to-Text, Text-to-Speech, Translation), Decision (Personalizer, Anomaly Detector, Content Moderator), and Azure OpenAI Service.

Q18₂ What is Azure Machine Learning?

Answer : Azure Machine Learning (Azure ML) is a cloud platform for building, training, deploying, and managing machine learning models at scale. Features: automated ML (AutoML) for non-experts, Designer (drag-and-drop ML pipeline builder), Notebooks (Jupyter), ML pipelines for automation, model registry, managed compute clusters for training, and model deployment to real-time endpoints or batch inference. Supports Python, R, and popular ML frameworks.

Q18₃ What is Azure OpenAI Service?

Answer : Azure OpenAI Service provides access to OpenAI's powerful language models (GPT-4, GPT-3.5, Codex, DALL-E, Embeddings) through Azure's infrastructure. Benefits over direct OpenAI API: enterprise security (VNet, private endpoints), regional compliance, content filtering, usage monitoring, and Azure AD integration. Used for building chatbots, code generation, summarization, translation, and custom generative AI applications.

Q18₄ What is Azure Bot Service?

Answer : Azure Bot Service is a platform for developing intelligent chatbots that interact naturally with users across multiple channels (Microsoft Teams, Slack, Facebook Messenger, SMS, web). It uses Azure AI Language for NLU and integrates with Azure OpenAI. Bot Framework SDK supports C# and Python. Power Virtual Agents (now Microsoft Copilot Studio) provides no-code bot creation for business users.

Q18₅ What is Azure Cognitive Search?

Answer : Azure Cognitive Search (formerly Azure Search) is an AI-powered cloud search service for building rich search experiences over private, heterogeneous content. Features: full-text search, semantic search (understand intent, not just keywords), AI enrichment (extract entities, sentiment, key phrases from documents), vector search (similarity search for AI applications), and faceted navigation. Used for e-commerce search, knowledge bases, and document discovery.

**Q18
6** What is Azure Synapse Analytics?

Answer : Azure Synapse Analytics is an integrated analytics service combining big data and data warehousing. Components: Synapse SQL (serverless and dedicated SQL pools for data warehouse queries), Apache Spark pools (big data processing), Synapse Pipelines (data integration and ETL), Power BI integration, Azure ML integration, and Synapse Link (real-time analytical data access from operational databases). Replaces Azure SQL Data Warehouse.

**Q18
7** What is Azure Databricks?

Answer : Azure Databricks is a fast, easy, and collaborative Apache Spark-based analytics platform optimized for Azure. It provides: interactive notebooks, managed Spark clusters, Delta Lake (reliable data lake), MLflow for ML lifecycle management, and Unity Catalog for data governance. Used for: big data processing, machine learning, real-time streaming analytics. Joint offering from Microsoft and Databricks with deep Azure integration.

**Q18
8** What is Power BI and its Azure integration?

Answer : Power BI is Microsoft's business intelligence and data visualization service. In Azure: Power BI integrates with Synapse Analytics, Azure SQL, Cosmos DB, Blob Storage, Event Hubs, and 100+ data sources. Power BI Embedded allows embedding reports in custom applications. Azure Analysis Services provides enterprise-grade semantic models. Power BI Premium on Azure provides AI-powered insights, large dataset support, and paginated reports.

**Q18
9** What is Azure Data Factory?

Answer : Azure Data Factory (ADF) is a cloud-based data integration and ETL (Extract, Transform, Load) service. It creates data pipelines that ingest data from 90+ sources, transform it, and load it to Azure Synapse, Blob Storage, SQL Database, or other destinations. ADF supports: code-free drag-and-drop pipeline design, triggers (schedule, event-based), data flows for transformation, and SSIS package execution for migration.

**Q19
0** What is Azure Stream Analytics?

Answer : Azure Stream Analytics is a real-time analytics service for processing streaming data from IoT devices, clickstreams, social media, and applications. It uses a SQL-like query language to analyze and transform data streams with millisecond latency. Common scenarios: real-time dashboards, anomaly detection, IoT telemetry processing, fraud detection, and fleet tracking. Integrates with Event Hubs, IoT Hub, and Blob Storage.

Q19₁ What is Azure Event Hubs?

Answer : Azure Event Hubs is a big data streaming platform and event ingestion service capable of receiving and processing millions of events per second. It acts as a front door for event pipelines — producers send events to Event Hubs, consumers (Stream Analytics, Functions, Logic Apps) process them. Features: Kafka compatibility (migrate Kafka workloads), capture to Blob/ADLS, geo-disaster recovery, schema registry.

Q19₂ What is Azure IoT Hub?

Answer : Azure IoT Hub is a managed service for bidirectional communication between IoT applications and millions of IoT devices. Features: device-to-cloud telemetry (send sensor data), cloud-to-device messaging (commands to devices), device management (update firmware), device twins (synchronize device state), and integration with Stream Analytics and Functions. IoT Hub is the backbone of Azure IoT solutions.

Q19₃ What is Azure IoT Central?

Answer : Azure IoT Central is a fully managed IoT SaaS platform that simplifies building and managing IoT solutions without infrastructure management. Unlike IoT Hub (which requires custom code), IoT Central provides a complete IoT application platform: device management, data visualization dashboards, rules-based alerts, and integration with Azure services. Ideal for organizations wanting a ready-to-use IoT platform with low development effort.

Q19₄ What is Azure Digital Twins?

Answer : Azure Digital Twins is a platform for creating digital representations (models) of real-world environments — factories, buildings, cities, supply chains. Twin graphs model relationships between things (room contains sensor, machine produces output). Real-time data from IoT devices updates the twin model. Use cases: smart buildings (optimize HVAC based on occupancy), factory digital twins for predictive maintenance, urban mobility simulation.

Q19₅ What is Azure Maps?

Answer : Azure Maps is a geospatial service providing APIs for maps, routing, spatial analytics, geocoding, and weather data. Features: interactive maps (built on Azure CDN), turn-by-turn routing, geofencing (trigger events when devices enter/exit areas), search (find addresses and places), traffic data, weather integration, and indoor mapping. Integrates with Power BI, Azure Functions, and IoT solutions.

**Q19
6** What is Azure Cognitive Services Language?

Answer : Azure AI Language (part of Cognitive Services) provides NLP capabilities: Text Analytics (sentiment analysis, key phrase extraction, language detection, named entity recognition), Azure OpenAI (GPT models), CLU (Conversational Language Understanding – replaces LUIS for building intent-based chatbots), Custom text classification, Question Answering (build FAQ chatbots from documents), and Translator (100+ language translation).

**Q19
7** What is Azure Computer Vision?

Answer : Azure Computer Vision is an AI service for extracting information from images and videos. Capabilities: Image Analysis (detect objects, brands, celebrities, generate image descriptions), OCR/Read (extract text from images and documents), Spatial Analysis (detect and count people in video, maintain social distancing analytics), Custom Vision (train custom image classification and object detection models with your own images).

**Q19
8** What is Azure Personalizer?

Answer : Azure Personalizer is a reinforcement learning-based AI service that personalizes content and experiences for individual users based on their behavior and context. Applications choose from a ranked list of actions; users interact with the top recommendation; Personalizer learns which action led to desired outcomes (clicks, purchases, time spent). Use cases: product recommendations, news personalization, UI layout optimization.

**Q19
9** What is Azure Anomaly Detector?

Answer : Azure Anomaly Detector is an AI service that identifies anomalies (unexpected data points) in time-series data without requiring ML expertise. It adapts automatically to your data patterns (seasonal trends, business cycles). Use cases: detecting equipment failures from IoT sensor data, identifying unusual traffic patterns, financial fraud detection, and monitoring application metrics for unexpected changes.

**Q20
0** What is Azure Form Recognizer (Document Intelligence)?

Answer : Azure AI Document Intelligence (formerly Form Recognizer) extracts structured data from documents using AI. Pre-built models: invoices, receipts, ID cards, business cards, W-2 forms, health insurance cards. Custom models: train on your specific form types. Use cases: automate data entry from invoices/receipts, digitize paper records, extract data from government forms. Outputs structured JSON with confidence scores.

Q20₁ What is the difference between AI Services, Azure ML, and Azure OpenAI?

Answer : Azure AI Services: pre-built AI APIs requiring no ML knowledge – use for standard tasks (translation, OCR, sentiment). Azure Machine Learning: platform for data scientists to build, train, and deploy custom ML models – when pre-built APIs don't fit your specific problem. Azure OpenAI Service: access to large language models (GPT-4, DALL-E) for generative AI applications – for text generation, code completion, creative content.

Q20₂ What is Azure Metrics Advisor?

Answer : Azure Metrics Advisor is an AI service for monitoring time-series metrics and diagnosing issues. It automatically detects anomalies, provides root cause analysis by correlating multiple metrics, and sends smart alerts. Unlike Anomaly Detector (single metric), Metrics Advisor handles multiple correlated metrics and helps identify which upstream metric caused the anomaly. Suitable for complex system monitoring.

Q20₃ What is Azure Health Bot?

Answer : Azure Health Bot is a cloud platform for healthcare organizations to build AI-powered virtual health assistants. It provides: built-in symptom checker, medical database integration, appointment scheduling, medication information, and FDA-approved clinical protocols. The Health Bot can be embedded in hospital websites, patient portals, and messaging platforms, helping patients find information without overloading call centers.

Q20₄ What is Responsible AI in Azure?

Answer : Microsoft's Responsible AI framework has six principles: Fairness (AI systems should treat all people equitably), Reliability and Safety (AI should perform reliably), Privacy and Security (AI should be secure and respect privacy), Inclusiveness (AI should empower everyone), Transparency (AI should be understandable), Accountability (humans should be accountable for AI systems). Azure provides Fairness tools, Interpretability, and Azure AI Content Safety.

Q20₅ What is Azure AI Content Safety?

Answer : Azure AI Content Safety is a service that detects harmful content in text and images – violence, hate speech, sexual content, self-harm. It provides multi-severity classifications (safe, low, medium, high) and can be integrated into applications to filter user-generated content. Used by social media platforms, gaming companies, and any application accepting user input to maintain safe online environments.

Q20
6 What is GitHub Copilot and Azure integration?

Answer : GitHub Copilot is an AI pair programmer powered by Azure OpenAI Codex that suggests code completions and entire functions in real time within VS Code, Visual Studio, and JetBrains IDEs. Azure integration: Copilot for Azure (preview) assists in Azure resource deployment and troubleshooting in the portal and CLI. Microsoft 365 Copilot integrates AI across Word, Excel, Outlook, and Teams using organizational data.

Q20
7 What is Azure AI Studio?

Answer : Azure AI Studio is a unified platform for building generative AI applications. It provides access to Azure OpenAI models, Azure AI Search for RAG (Retrieval Augmented Generation), evaluation tools, prompt flow for orchestrating AI workflows, and safety guardrails. Developers can build, test, and deploy AI applications end-to-end. AI Studio integrates with Azure ML for model fine-tuning and deployment.

Q20
8 What is Azure PlayFab?

Answer : Azure PlayFab is a complete backend platform for live games. Features: player authentication (social login, accounts), leaderboards, cloud saves, economy (virtual currency, items), matchmaking, analytics, LiveOps (events, A/B testing), push notifications, and multiplayer server hosting. PlayFab helps game developers add server-side features without building custom infrastructure, scaling to millions of players globally.

Q20
9 What is Azure Communication Services?

Answer : Azure Communication Services (ACS) enables developers to add real-time communication features to their applications: voice and video calling, chat, SMS, email, and Teams interoperability. ACS uses the same infrastructure as Microsoft Teams. Use cases: telemedicine (video consultations), customer support platforms, virtual events, and embedding communication in business apps without managing telephony infrastructure.

Q21
0 What is Azure Space?

Answer : Azure Space provides cloud computing capabilities for space and satellite applications. Services include: Azure Orbital Ground Station (rent ground station antennas for satellite communication), Satellite Connectivity (use satellites for remote area connectivity), and space data processing (process satellite imagery and telemetry in Azure). Partners with SpaceX Starlink, SES, Telesat for global connectivity solutions.

Section 8: Azure Databases

Questions 211 – 240

Q21
1 What is Azure SQL Database?

Answer : Azure SQL Database is a fully managed relational database service based on the latest stable version of Microsoft SQL Server. It's a PaaS offering — Microsoft handles OS patching, backups, high availability, and scaling. Features: serverless (auto-pause when inactive), elastic pools (share resources among multiple DBs), Hyperscale (up to 100 TB), geo-replication, auto-failover groups, built-in intelligence (automatic index tuning), and 99.99% SLA.

Q21
2 What is Azure SQL Managed Instance?

Answer : Azure SQL Managed Instance (SQL MI) is a PaaS service providing near 100% compatibility with on-premises SQL Server. Unlike SQL Database (which is a re-architected service), SQL MI supports SQL Server features not available in SQL Database: SQL Agent, Service Broker, linked servers, cross-database queries, CLR, Database Mail. Ideal for 'lift and shift' migrations of SQL Server workloads requiring full SQL Server engine compatibility.

Q21
3 What is the difference between SQL Database and SQL Managed Instance?

Answer : SQL Database: single database or elastic pool, cloud-native design, some SQL Server features not supported, vCore or DTU purchasing models, lower cost for simple databases. SQL Managed Instance: full SQL Server instance in the cloud, supports 99% of SQL Server features including SQL Agent/CLR/linked servers, VNet deployed (no public endpoint by default), higher cost. Choose MI for compatibility, SQL DB for modern cloud-native apps.

Q21
4 What is Azure Database for MySQL?

Answer : Azure Database for MySQL is a fully managed relational database service based on the MySQL Community Edition. Microsoft manages: hardware, OS, database software patching, backups, high availability, and monitoring. Features: up to 16 TB storage, 99.99% SLA with zone-redundant HA, read replicas for scaling reads, point-in-time restore (up to 35 days), and geo-restore for regional disaster recovery.

Q21
5 What is Azure Database for PostgreSQL?

Answer : Azure Database for PostgreSQL is a fully managed relational database service for PostgreSQL. Deployment options: Single Server (legacy, being retired), Flexible Server (recommended – zone redundancy, burstable compute, granular maintenance window control). Features: Citus extension for distributed/hyperscale PostgreSQL, pgvector support for AI/vector embeddings, online migrations, and built-in connection pooling.

**Q21
6** What is Azure Cosmos DB?

Answer : Azure Cosmos DB is Microsoft's fully managed, globally distributed, multi-model NoSQL database service. Key features: millisecond response times at any scale, multiple consistency levels (from eventual to strong), turnkey global distribution (multi-region writes), 99.999% SLA for multi-region writes, multiple APIs: SQL (Core), MongoDB, Cassandra, Gremlin (graph), Table. Billed by Request Units (RU/s) consumed.

**Q21
7** What are Cosmos DB consistency levels?

Answer : Cosmos DB offers 5 consistency levels with different tradeoffs between consistency and performance/cost: Strong (highest consistency, reads always reflect latest write – highest latency/cost). Bounded Staleness (reads lag writes by configurable amount). Session (consistent reads within a session – default, most commonly used). Consistent Prefix (reads never see out-of-order writes). Eventual (lowest consistency, highest performance/lowest cost).

**Q21
8** What is Azure Cache for Redis?

Answer : Azure Cache for Redis is a fully managed, in-memory cache based on the open-source Redis (Remote Dictionary Server) database. It dramatically improves application performance by caching frequently accessed data in memory (microsecond latency) instead of querying slow databases. Use cases: session caching, full-page caching, distributed lock management, real-time leaderboards, message broker (pub/sub). Tiers: Basic, Standard, Premium, Enterprise.

**Q21
9** What is Azure SQL Elastic Pool?

Answer : Azure SQL Elastic Pool allows multiple databases to share a fixed amount of compute resources (eDTUs or vCores) from a pool, at a price lower than provisioning individual databases. Ideal for SaaS applications with multiple tenant databases with unpredictable, variable usage patterns – inactive databases don't waste allocated resources while active ones burst using the shared pool capacity.

**Q22
0** What is Azure Database Migration Service?

Answer : Azure Database Migration Service (DMS) is a fully managed service for migrating databases from on-premises or cloud to Azure with minimal downtime. Supports: SQL Server to SQL Database/MI, MySQL to Azure Database for MySQL, PostgreSQL to Azure Database for PostgreSQL, Oracle to SQL Database/MI, and MongoDB to Cosmos DB. DMS performs schema conversion, data migration, and monitors the process.

Q22₁ What is Azure Synapse Link?

Answer : Azure Synapse Link is a cloud-native hybrid transactional and analytical processing (HTAP) capability that automatically syncs operational data from Azure Cosmos DB, Azure SQL Database, and SQL Server to Azure Synapse Analytics in near real time — without ETL pipelines or impacting operational database performance. This enables real-time analytical queries on live operational data.

Q22₂ What is Azure SQL Serverless?

Answer : Azure SQL Serverless is a compute tier for Azure SQL Database that automatically pauses the database when inactive and resumes when a connection is made. Billing is per second of compute used when active — no charges when paused (storage is still billed). Ideal for databases with intermittent, unpredictable usage (dev/test, infrequently used apps). Includes auto-scaling within configured min/max vCore range.

Q22₃ What is Azure SQL Hyperscale?

Answer : Azure SQL Hyperscale is a cloud-native architecture for Azure SQL Database that decouples compute from storage, enabling rapid scaling and storage up to 100 TB. Page servers provide distributed storage; named replicas provide read scale-out. Database creation and scaling operations complete in minutes regardless of data size. Backups are near-instantaneous via storage snapshots. Supports up to 5 readable replicas.

Q22₄ What is Azure Table Storage vs Cosmos DB Table API?

Answer : Azure Table Storage: simple, low-cost key-value store within Azure Storage. Limited query capabilities, single region. Cosmos DB Table API: drop-in replacement with the same table model but adds global distribution, multi-region writes, single-digit millisecond latency, SLA guarantees, and better indexing. Migrating from Table Storage to Cosmos DB Table API requires minimal code changes (same SDK) but provides dramatically better performance and features.

Q22₅ What is the Cosmos DB Request Unit (RU)?

Answer : A Request Unit (RU) is the currency for Azure Cosmos DB throughput. All database operations (reads, writes, queries) consume RUs based on complexity. One RU = reading a 1 KB item by its ID (cheapest operation). Writes and complex queries consume more RUs. You provision RU/s (throughput) in advance or use serverless (pay per RU consumed). RUs abstract compute, memory, and I/O into a single metric for capacity planning.

**Q22
6** What is Azure MariaDB?

Answer : Azure Database for MariaDB is a fully managed database service based on MariaDB Community Edition (a MySQL fork). Microsoft managed service providing automatic backups, OS patching, monitoring, and HA. However, Microsoft announced retirement of Azure Database for MariaDB — existing customers are encouraged to migrate to Azure Database for MySQL Flexible Server, which has broader feature support and longer-term investment.

**Q22
7** What is data residency in Azure databases?

Answer : Data residency ensures that data is stored and processed only within a specified geographic region — required for compliance with regulations like GDPR, Australia Privacy Act, and Brazil LGPD. Azure databases let you choose the region for deployment. Azure SQL Database, Cosmos DB, and MySQL/PostgreSQL store data within your chosen region and paired region (for geo-redundancy) unless you opt out of geo-replication.

**Q22
8** What is Azure SQL Always Encrypted?

Answer : Always Encrypted is a SQL Server and Azure SQL Database feature that protects sensitive data (SSNs, credit card numbers) by encrypting it on the client side before sending to the database. The database engine never sees plaintext data — even database administrators can't read encrypted columns. Encryption keys are stored in Azure Key Vault or Windows Certificate Store. Data is decrypted only in the client application.

**Q22
9** What is Azure Database for PostgreSQL Flexible Server?

Answer : Azure Database for PostgreSQL Flexible Server is the recommended deployment option providing: zone-redundant high availability with automatic failover, a flexible maintenance window (choose when updates occur), burstable compute for cost savings during low usage, major version upgrade support, and Citus extension for distributed PostgreSQL (sharding across nodes). Replaces the older Single Server option.

**Q23
0** What is Azure SQL Auto-Failover Groups?

Answer : Auto-failover groups are an Azure SQL Database and Managed Instance feature that manages geo-replication and failover for a group of databases to a secondary region. They provide: automatic failover during outages (or manual failover for maintenance), read/write listener endpoint (automatically points to current primary), read-only listener endpoint (for read workloads on secondary), and transparent application connection string after failover.

Q23 1 What is in-memory OLTP in Azure SQL?

Answer : In-Memory OLTP (Online Transaction Processing) is an Azure SQL Database feature that stores tables and stored procedures in memory-optimized format for dramatically faster transaction processing. Memory-optimized tables use a latch-free, optimistic concurrency model. Up to 30x faster than disk-based tables for write-heavy OLTP workloads. Available in Business Critical and Premium tiers. Ideal for high-frequency, low-latency transaction processing.

Q23 2 What is Azure Database for MySQL vs MySQL on a VM?

Answer : Azure Database for MySQL: PaaS – Microsoft manages OS, MySQL patches, backups, HA, storage scaling. No SSH access. Easier to manage. SLA-backed. Azure-specific features (PITR, geo-restore). MySQL on Azure VM: IaaS – you install and manage MySQL, OS, and hardware. Full SSH access and configuration control. Supports all MySQL features and plugins. Use PaaS for most workloads; use VM only when you need configurations not supported by PaaS.

Q23 3 What is Azure Cosmos DB for MongoDB?

Answer : Cosmos DB for MongoDB is an API layer that makes Cosmos DB behave like a MongoDB database. MongoDB drivers, tools (Compass, mongosh), and applications can connect to Cosmos DB without code changes – just change the connection string. Benefits: Cosmos DB's global distribution, multi-region writes, 5 consistency levels, and unlimited scale – with MongoDB wire protocol compatibility. No need to run or manage MongoDB servers.

Q23 4 What is read replicas in Azure databases?

Answer : Read replicas asynchronously replicate data from a primary database server to one or more read-only replica servers. This offloads read queries (reports, analytics, BI) from the primary server, reducing its load and improving performance. Available in: Azure Database for MySQL (up to 5 replicas), PostgreSQL Flexible Server (up to 5 replicas), and SQL Database/MI (geo-secondary). Data on replicas may lag slightly (eventual consistency).

Q23 5 What is Azure SQL DTU vs vCore?

Answer : DTU (Database Transaction Unit): older purchasing model bundling CPU, memory, and I/O into a single metric. Simpler but less flexible. vCore (virtual Core): newer model allowing independent selection of compute (cores), memory, and storage. More transparent, supports Azure Hybrid Benefit, better for migrating from on-premises SQL Server. Microsoft recommends vCore for most new deployments. Serverless is available only with vCore model.

Q23
6 What is point-in-time restore (PITR)?

Answer : Point-in-time restore allows recovering a database to any specific moment within the backup retention period. Azure SQL Database retains 1-35 days of backups (configurable). MySQL and PostgreSQL retain up to 35 days. PITR creates a new database from the backed-up data at the specified timestamp. Useful for recovering from accidental data deletion, incorrect updates, or application bugs that corrupted data.

Q23
7 What is Azure SQL Geo-replication?

Answer : Azure SQL Database active geo-replication replicates a database asynchronously to up to 4 readable secondary databases in different regions. Secondaries can serve read traffic and be failed over to within seconds. Unlike auto-failover groups, geo-replication doesn't provide automatic failover — you must manually initiate failover. Use auto-failover groups for automatic failover; use geo-replication for more control and readable secondaries.

Q23
8 What is Cosmos DB Global Distribution?

Answer : Cosmos DB's global distribution allows you to add and remove Azure regions to your database account at any time with a single click or API call. Data is automatically replicated across all configured regions. With multi-region writes enabled, users write to the nearest region with single-digit millisecond latency. Conflict resolution policies (Last Write Wins, Custom) handle simultaneous writes to different regions.

Q23
9 What is Azure SQL Edge?

Answer : Azure SQL Edge is an optimized relational database engine for IoT and edge computing scenarios. It runs on small, low-power devices (ARM64 and x64) and provides SQL Server engine capabilities — T-SQL, RBAC, encryption — plus built-in AI/ML capabilities for real-time analytics on IoT streaming data. Azure SQL Edge uses the same engine as SQL Server, so development skills transfer directly.

Q24
0 What is the Cosmos DB serverless tier?

Answer : Cosmos DB serverless is a consumption-based model where you pay only for the Request Units consumed by your operations — no provisioned throughput required. Ideal for: development and testing, applications with intermittent traffic, new applications where usage patterns are unknown. No minimum cost when idle. Serverless containers have limits (currently 1 TB storage, 5,000 RU/s burst). Not suitable for sustained high-throughput production workloads.

Section 9: Azure DevOps & Developer Tools

Questions 241 – 270

Q24
1 What is Azure DevOps?

Answer : Azure DevOps is a collection of development tools for planning, developing, testing, and delivering software. Services: Azure Boards (Agile project management — backlogs, sprints, Kanban), Azure Repos (Git repositories), Azure Pipelines (CI/CD — build, test, deploy), Azure Test Plans (manual and exploratory testing), Azure Artifacts (package management — NuGet, npm, Maven, Python). DevOps is available as SaaS (Azure DevOps Services) or on-premises (Azure DevOps Server).

Q24
2 What is Azure Pipelines?

Answer : Azure Pipelines is a CI/CD (Continuous Integration/Continuous Delivery) service that automatically builds, tests, and deploys code changes. Supports: any language (Python, Java, .NET, Node.js, Go), any platform, any cloud or on-premises. Pipeline defined in YAML (azure-pipelines.yml) in the repository. Features: parallel jobs, matrix builds, environment approvals, release gates, integration with GitHub and Azure Repos.

Q24
3 What is Azure Boards?

Answer : Azure Boards is an Agile project management tool for planning and tracking work. Supports: Scrum (sprints, sprint planning, velocity), Kanban (visual workflow boards), Agile (stories, tasks, bugs), and CMMI methodologies. Features: customizable work item types, hierarchical backlogs (Epics > Features > User Stories > Tasks), dashboards and widgets, GitHub integration (link commits and PRs to work items), and reporting.

Q24
4 What is Azure Repos?

Answer : Azure Repos provides unlimited free private Git repositories (or TFVC — Team Foundation Version Control) with code review, pull requests, and branch policies. Features: branch protection policies (require code review, passing builds, linked work items), pull request templates, inline commenting, search, and integration with Azure Pipelines for CI/CD. Supports any Git client, GitHub Desktop, and VS Code.

Q24
5 What is Azure Artifacts?

Answer : Azure Artifacts is a package management service for storing and sharing software packages: NuGet (.NET), npm (JavaScript), Maven (Java), Python (pip), and Universal Packages (any file type). It provides private package feeds your team can publish to and consume from. Upstream sources allow proxying public registries (npmjs.com, nuget.org) through a private feed, with packages cached for reliability.

Q24
6

What is GitHub Actions vs Azure Pipelines?

Answer
:

GitHub Actions: native CI/CD for GitHub repositories, tightly integrated with GitHub events (push, PR, issue), large community marketplace of pre-built Actions, free for public repos. Azure Pipelines: more mature, enterprise features (release stages, environments, deployment gates), supports multiple repositories (GitHub, Bitbucket, Azure Repos, SVN), better for complex enterprise workflows. Both use YAML, both run on Microsoft-hosted agents.

Q24
7

What is Azure DevTest Labs?

Answer
:

Azure DevTest Labs lets developers and testers quickly create environments in Azure while minimizing waste and cost. Features: self-service VM creation from pre-approved templates and images, automatic shutdown schedules to reduce costs, spending policies (limit VMs per user, max cost per lab), artifact libraries (scripts that install tools), and integration with Azure DevOps pipelines for automated environment provisioning.

Q24
8

What is Visual Studio Code and Azure extensions?

Answer
:

Visual Studio Code (VS Code) is Microsoft's free, open-source, cross-platform code editor. Azure extensions add Azure management capabilities directly in VS Code: Azure Account (sign in and manage subscriptions), Azure Resources (browse and manage Azure resources), Azure Functions (create, run, debug, deploy Functions), Azure App Service (deploy web apps), Azure Storage, Kubernetes, Docker, and Azure Databases extensions.

Q24
9

What is Azure App Service deployment slots?

Answer
:

Deployment slots are live environments within an Azure App Service. The default slot is 'production.' You can create additional slots (staging, testing) to deploy new versions without affecting production. Slots have their own hostnames and can be independently deployed. 'Swap' operation exchanges slots instantly — routing traffic from staging to production with zero downtime. If a new version has issues, swap back to restore previous version immediately.

Q25
0

What is Infrastructure as Code (IaC)?

Answer
:

Infrastructure as Code (IaC) is the practice of managing and provisioning infrastructure through machine-readable configuration files rather than manual processes. Benefits: version control (track infrastructure changes in Git), consistency (same config produces same infrastructure), repeatability (deploy identical environments), automation (integrate with CI/CD), documentation (code IS the documentation). Azure IaC tools: ARM templates, Bicep, Terraform, Pulumi.

Q25
1

What is Azure DevOps vs GitHub?

Answer
:

GitHub: primarily source code hosting with integrated CI/CD (GitHub Actions), strong open-source community, Issues for bug tracking, GitHub Packages for packages. Azure DevOps: complete ALM (Application Lifecycle Management) suite with Boards, Repos, Pipelines, Artifacts, Test Plans – designed for enterprise teams. Microsoft owns both. Recommendation: GitHub for new projects and open-source; Azure DevOps when you need full Boards/Test Plans integration.

Q25
2

What is continuous integration (CI)?

Answer
:

Continuous Integration is the practice of automatically building and testing code changes every time a developer commits to the repository. CI ensures that: code compiles, automated tests pass, code quality meets standards (linting, static analysis). Benefits: detect integration issues early (when they're cheap to fix), maintain always-releasable code, provide fast feedback to developers. Azure Pipelines and GitHub Actions implement CI.

Q25
3

What is continuous delivery (CD)?

Answer
:

Continuous Delivery extends CI by automatically deploying tested code changes to staging or production environments. With CD, every change that passes CI tests is a release candidate. Continuous Delivery: automates deployment but may require manual approval for production. Continuous Deployment: fully automated – every passing change goes directly to production without human intervention. Azure Pipelines environments and approval gates implement CD.

Q25
4

What is Azure Container Registry (ACR) in DevOps?

Answer
:

In DevOps workflows, ACR is used to store container images built by CI/CD pipelines. Pipeline: code commit → build Docker image → push to ACR → deploy to AKS/ACI from ACR. ACR Tasks automate image builds on code commit or base image update. ACR integrates with Azure AD for authentication and RBAC for access control. Geo-replication distributes images close to deployment regions for faster pulls.

Q25
5

What is GitOps?

Answer
:

GitOps is a DevOps practice using Git as the single source of truth for declarative infrastructure and application configuration. Changes to infrastructure are made through Git commits and pull requests – automated systems detect changes and apply them to the cluster. For Kubernetes: tools like Flux and Argo CD watch Git repositories and automatically sync the cluster state. Azure Arc and AKS support GitOps with Flux.

Q25
6

What is Azure Monitor Application Insights in DevOps?

Answer : Application Insights integrates into DevOps workflows for continuous monitoring. Features: live metrics during deployment (monitor failure rate as new version rolls out), availability tests (ping tests to verify app is running), release annotations (marks deployment events on charts – correlate performance changes with deployments), smart detection (automatically alerts on anomalous failure rates after deployment).

Q25
7 What are Azure Pipelines agents?

Answer : Pipeline agents execute the pipeline jobs. Microsoft-hosted agents: Azure provides VMs (Windows, Ubuntu, macOS) preconfigured with common tools – no maintenance required, fresh VM per job. Self-hosted agents: you provide and manage VMs, containers, or physical machines – full control, custom software, access to on-premises resources, persistent storage between jobs. For build jobs needing internet access, Microsoft-hosted agents are recommended.

Q25
8 What is Azure Static Web Apps?

Answer : Azure Static Web Apps automatically builds and deploys full-stack web applications from GitHub or Azure DevOps. It hosts: static frontend (HTML, CSS, JS, React, Angular, Vue, Blazor) globally via CDN, serverless API backend (Azure Functions), built-in auth (GitHub, Twitter, Microsoft, Google, custom OpenID Connect), free SSL certificate, custom domains, and preview environments for pull requests (staging URLs auto-created).

Q25
9 What is the Azure SDK?

Answer : Azure SDKs are language-specific libraries for interacting with Azure services programmatically from your application code. Available for: Python, .NET/C#, Java, JavaScript/TypeScript, Go, C++, Ruby. SDKs handle authentication (Azure AD, connection strings), retry logic, serialization, and service-specific abstractions. The Azure SDK Design Guidelines ensure consistency across languages. Install via pip, NuGet, Maven, npm, etc.

Q26
0 What is Azure API Management (APIM)?

Answer : Azure API Management is a hybrid, multi-cloud platform for managing APIs across all environments. Features: API gateway (proxies and routes API calls), developer portal (documentation, interactive testing, subscription management), policies (transform/validate requests without code changes – add headers, rate-limit, cache responses, OAuth validation), analytics, and monetization. Supports backends in Azure, on-premises, and other clouds.

Q26
1 What is Azure Service Bus?

Answer : Azure Service Bus is an enterprise message broker with queues (one-to-one, at-least-once delivery) and topics/subscriptions (one-to-many, pub/sub model). It provides: FIFO guaranteed delivery, dead-letter queues, message sessions, transactions, duplicate detection, and scheduled delivery. Compared to Azure Queue Storage: Service Bus offers richer features (ordering, transactions, 256 KB messages, topics) while Queue Storage is simpler and cheaper.

**Q26
2** What is Azure Event Grid?

Answer : Azure Event Grid is a serverless event routing service that connects event sources to event handlers using publish-subscribe model. Event sources: Azure services (Blob Storage, Resource Manager, IoT Hub, ACR) or custom sources. Event handlers: Azure Functions, Logic Apps, Event Hubs, Service Bus, webhooks. Event Grid filters events by type, subject, or data. Scales automatically – millions of events per second at low cost.

**Q26
3** What is the difference between Service Bus, Event Grid, and Event Hubs?

Answer : Event Grid: event-driven serverless routing, discrete events (something happened), fan-out to multiple subscribers, small payloads, at-most-once delivery. Service Bus: enterprise messaging, guaranteed ordered delivery, competing consumers, transactions, large messages. Event Hubs: high-throughput streaming data ingestion (millions/second), time-series data, multiple consumers replaying streams, big data pipeline integration. Use: Event Grid for reactive apps, Service Bus for reliable messaging, Event Hubs for big data streaming.

**Q26
4** What is Azure Notification Hubs?

Answer : Azure Notification Hubs provides a scalable, cross-platform push notification engine for mobile applications. It abstracts the complexity of platform-specific push services (APNs for iOS, FCM for Android, WNS for Windows). Benefits: send notifications to millions of devices from a single API call, segment audiences with tags and templates, per-platform customization, and telemetry on notification delivery. Decouples push notification logic from app backends.

**Q26
5** What is Azure SignalR Service?

Answer : Azure SignalR Service provides a fully managed service for adding real-time web functionality to applications. It handles WebSocket connections at scale, enabling servers to push content to clients instantly without polling. Use cases: live dashboards, collaborative apps (online document editing), gaming (multiplayer), chat applications, live sports/auction updates. Integrates with Azure Functions (serverless SignalR) for event-driven real-time features.

**Q26
6** What is Azure Logic Apps Standard vs Consumption?

Answer : Logic Apps Consumption: multi-tenant, serverless, pay per action execution, easy to use, hosted by Microsoft. Logic Apps Standard: single-tenant, runs on App Service or container, supports stateful and stateless workflows, VNET integration, local development with VS Code, can run multiple workflows per instance (lower cost for high volume). Standard provides more control, better performance, and workflow isolation vs the shared Consumption tier.

Q26
7 What is Azure App Configuration?

Answer : Azure App Configuration is a centralized service for managing application settings and feature flags. Instead of hardcoding configuration in app code or multiple config files, all configuration is stored centrally and fetched at runtime. Features: key-value store with hierarchical naming, configuration versioning and snapshots, feature flags (enable/disable features without redeployment), dynamic configuration refresh, and integration with Key Vault for secrets.

Q26
8 What is Azure Relay?

Answer : Azure Relay enables secure communication between on-premises services and cloud applications without opening inbound firewall ports. The on-premises service creates an outbound connection to Azure Relay; cloud services connect through Relay to reach on-premises. Two types: Hybrid Connections (WebSockets — any language, any platform), WCF Relay (Windows Communication Foundation). Useful for exposing on-premises APIs or services securely without VPN.

Q26
9 What is Azure Service Fabric Mesh?

Answer : Azure Service Fabric Mesh (now in maintenance mode) was a serverless managed platform for deploying Service Fabric microservices applications without managing infrastructure. Developers defined applications in YAML/JSON; Azure provisioned and managed the underlying Service Fabric cluster. Microsoft has shifted focus to AKS for containerized microservices deployment and recommends evaluating AKS for new microservices workloads.

Q27
0 What is GitHub Advanced Security for Azure DevOps?

Answer : GitHub Advanced Security for Azure DevOps brings GitHub's code scanning capabilities to Azure Repos: Secret Scanning (detect accidentally committed API keys, tokens, passwords), Dependency Scanning (identify vulnerabilities in open-source dependencies), and Code Scanning with CodeQL (find security vulnerabilities in custom code). Results appear in Azure DevOps with actionable remediation guidance, helping teams ship more secure code.

Section 10: Azure Pricing, Support & Exam Prep

Questions 271 – 300

Q27
1 What are the Azure support plans?

Answer : Microsoft offers 5 Azure support plans: Basic (free — documentation, self-help, Azure Advisor, Service Health dashboard, no technical support). Developer (\$29/month — business hours email, 8-hour severity C response). Standard (\$100/month — 24/7 email/phone, 2-hour sev B, 4-hour sev C). Professional Direct (\$1,000/month — 1-hour sev A, proactive guidance, architecture support). Premier/Unified (enterprise — dedicated TAM, custom SLA, Mission Critical support).

Q27
2 What is Azure Service Health used for?

Answer : Azure Service Health provides personalized information about service issues, planned maintenance, and health advisories that affect your specific subscriptions and regions. Three components: Azure Status (global outage page), Service Health (your subscriptions/regions), Resource Health (specific resource health — 'Is my VM running?'). You can configure alerts to get notified via email, SMS, webhooks, or Azure Monitor action groups when issues occur.

Q27
3 How does Azure billing work?

Answer : Azure uses a monthly billing cycle. Resources are metered by usage: compute (per second/minute/hour), storage (GB-months), data transfer (GB out to internet), and API calls (per million). Azure generates an invoice monthly showing itemized charges. Payment methods: credit card (pay-as-you-go), invoice (NET 30 — for enterprises). Enterprise Agreement (EA) customers prepay for Azure usage in Azure Prepayment (formerly Monetary Commitment) and consume from that credit.

Q27
4 What is an Enterprise Agreement (EA)?

Answer : An Azure Enterprise Agreement is a volume licensing agreement for organizations with 500+ users or devices. Benefits: significant discounts (up to 60%) compared to PAYG, Azure Prepayment (upfront commitment for predictable spend), centralized billing and department cost tracking, extended payment terms, Azure Hybrid Benefit included. EA portal provides detailed usage reporting and cost management across departments.

Q27
5 What is a Cloud Solution Provider (CSP)?

Answer : Cloud Solution Providers are Microsoft partners authorized to sell Azure subscriptions and manage billing on behalf of customers. Benefits for customers: consolidated billing (Azure + Microsoft 365 on one bill), support from local partner, partner-added services and customization. CSP partners can provide support and have full access to customer tenant. Pricing may be different from PAYG – can be higher or lower depending on partner.

**Q27
6** What is the Azure free tier?

Answer : Azure free tier includes: 12 months of popular services free (VMs, databases, storage), always-free services with monthly limits (Azure Functions 1M executions, Cosmos DB 1,000 RU/s, Blob Storage 5 GB), and \$200 credit for 30 days. After free period: you must upgrade to Pay-As-You-Go to continue using services. Free account requires credit card for identity verification but charges don't occur automatically without upgrade.

**Q27
7** How can you reduce Azure costs?

Answer : Cost optimization strategies: Reserved Instances (commit 1-3 years for up to 72% savings). Spot VMs (use excess capacity at up to 90% discount). Azure Hybrid Benefit (use existing licenses). Right-sizing (use Azure Advisor to identify underutilized resources). Auto-shutdown (stop dev/test VMs after hours). Lifecycle management (move storage to cheaper tiers). Delete unused resources. Azure Cost Management + Billing for visibility and budgets.

**Q27
8** What is Azure Cost Management + Billing?

Answer : Azure Cost Management + Billing is a suite of tools for monitoring, allocating, and optimizing Azure costs: Cost Analysis (interactive visualizations of spending by service, region, tag, resource group), Budgets (set spending thresholds with email/webhook alerts), Cost Alerts (budget, anomaly, and department spending alerts), Recommendations (Advisor integration), Exports (schedule regular cost reports to storage), and Azure Marketplace purchase management.

**Q27
9** What is a spending limit in Azure?

Answer : Azure spending limits prevent charges beyond a set amount for accounts with monthly credits (Visual Studio, MSDN, free trial). When the limit is reached, Azure suspends services for the rest of the billing period. Note: spending limits are NOT available for Pay-As-You-Go or Enterprise Agreement customers – only for credit-based subscriptions. Azure Budgets provide budget alerts but don't hard-stop spending like spending limits.

**Q28
0** What are Azure quotas and limits?

Answer : Azure quotas are default limits on the number of resources per subscription per region: 25,000 VMs per region, 250 storage accounts per subscription, 250 VNets per region, etc. These are soft limits that can be increased by submitting a support request. Hard limits cannot be increased (e.g., max 5 resource groups with Locks per subscription). Check quotas in Azure Portal under Subscriptions > Usage + quotas.

Q28
1 What is the Microsoft Customer Agreement (MCA)?

Answer : The Microsoft Customer Agreement (MCA) is a simplified purchase agreement replacing the old Microsoft Online Subscription Agreement (MOSA) for individual Azure subscriptions. MCA provides: digital-first acceptance (no paper), billing accounts and profiles for cost management, invoice sections for departmental chargebacks, and the ability to transfer subscriptions between billing accounts. MCA supports all purchasing channels (direct, CSP, EA).

Q28
2 What are Azure region pricing differences?

Answer : Azure pricing varies by region due to different local costs (electricity, real estate, labor, taxes). East US is typically one of the cheapest regions. Europe and Asia Pacific regions generally cost 10-15% more. Some regions (Brazil South, South Africa) cost significantly more. When cost is a priority, deploy in regions like East US, West US, or North Europe. Check the pricing calculator for region-specific pricing.

Q28
3 What is Azure data transfer pricing?

Answer : Azure charges for some data transfers: Inbound (data going INTO Azure) is generally free. Outbound (data going FROM Azure to internet) is charged by GB – first 100 GB/month free, then \$0.087/GB (East US). Data transfer between Azure regions (cross-region) is charged at \$0.02/GB or higher. Data transfer within the same region is free. Data transfer between VNets in the same region via peering may have small costs.

Q28
4 What is Azure Reserved Capacity for databases?

Answer : Similar to Reserved VM Instances, Azure offers reserved capacity for database services: Azure SQL Database, SQL Managed Instance, Cosmos DB, MySQL, PostgreSQL, Redis Cache, Synapse Analytics. Committing to 1 or 3 years provides up to 80% savings compared to PAYG. Reservations are applied to matching database resources in the same region and subscription. Reservations can be exchanged or refunded (with early termination fees).

Q28
5 What is Azure Dev/Test pricing?

Answer : Azure Dev/Test pricing provides significantly discounted rates for development and testing workloads. Requirements: active Visual Studio subscription. Benefits: no Windows VM licensing charges (Linux-equivalent pricing for Windows VMs), discounted rates on other services. Available through Azure Dev/Test subscriptions (individual VS subscribers) and Enterprise Dev/Test offer (for Enterprise Agreement customers). Dev/Test resources cannot run production workloads.

Q28
6 What compliance certifications does Azure have?

Answer : Azure has the largest compliance portfolio in the cloud: Global certifications: ISO 27001, ISO 27018, SOC 1/2/3, CSA STAR. US Government: FedRAMP High, DoD Impact Levels 2/4/5, CJIS, ITAR. Industry: PCI DSS Level 1, HIPAA/HITECH, GxP (pharma). Regional: GDPR (EU), UK G-Cloud, Australia IRAP, Singapore MTCS, Germany C5, Japan CS Gold Mark. Total: 100+ compliance certifications across 50+ regions.

Q28
7 What is the Trust Center?

Answer : The Microsoft Trust Center (microsoft.com/trust-center) is a central resource for security, privacy, and compliance information about Microsoft cloud services. It provides: compliance documentation (audit reports, certifications), privacy policies, data protection information, security practices, and regional compliance requirements. The Trust Center includes the Service Trust Portal where you can download audit reports and compliance documents.

Q28
8 What is GDPR and Azure compliance?

Answer : GDPR (General Data Protection Regulation) is the EU's data protection regulation. Azure compliance: Microsoft is a GDPR-compliant data processor. Azure's Data Processing Agreement (DPA) includes GDPR commitments. Azure provides tools for compliance: Azure AD for consent management, Azure Policy for data residency, Microsoft Purview for data discovery and classification, Customer Lockbox for access control, and comprehensive audit logs.

Q28
9 What is Customer Lockbox?

Answer : Customer Lockbox provides a workflow for Microsoft support engineers to request access to customer data when needed for a support case. With Customer Lockbox enabled, Microsoft cannot access your data without your explicit approval. When an engineer needs access, you receive a request notification with details (what data, why, who) and must approve or deny it within a time limit. All decisions are logged for compliance audit purposes.

Q29
0 What is Azure Sovereign Cloud?

Answer : Azure Sovereign Clouds are physically and logically isolated instances of Azure designed for government and regulated industries with strict data sovereignty requirements. Types: Azure Government (US government agencies — FedRAMP High, DISA, CJIS), Azure China (operated by 21Vianet — complies with Chinese regulations), Azure Germany (data trustee model). Sovereign clouds are separate from global Azure and have different service availability.

Q29 1 What is the purpose of the Az-900 exam?

Answer : The Az-900: Microsoft Azure Fundamentals exam validates foundational knowledge of cloud concepts and Azure services — no technical experience required. It's ideal for: individuals beginning their Azure career, non-technical professionals (sales, purchasing, project managers) working with Azure teams, and people validating their cloud fundamentals before taking role-based certifications (Az-104, Az-204, etc.). The exam covers cloud concepts, Azure services, management, and governance.

Q29 2 What is the Microsoft Certification path after Az-900?

Answer : After Az-900 (Fundamentals), career paths: Administrator: AZ-104 Microsoft Azure Administrator. Developer: AZ-204 Developing Solutions for Azure. Architect: AZ-305 Designing Azure Solutions (requires AZ-104 first). Security: AZ-500 Azure Security Engineer. AI: AI-102 Azure AI Engineer. Data: DP-900 Azure Data Fundamentals or DP-203 Azure Data Engineer. DevOps: AZ-400 Azure DevOps Engineer. Choose path based on your job role.

Q29 3 What are the key Azure free services for Az-900 study?

Answer : For Az-900 exam preparation, use these free Azure resources: Azure free account (\$200 credit + 12 months free services), Microsoft Learn (free interactive modules specifically for Az-900 — learn.microsoft.com/certifications/azure-fundamentals), Microsoft Azure Documentation (docs.microsoft.com), Azure Quickstart Templates (GitHub), Azure Pricing Calculator, Azure TCO Calculator, and Azure Architecture Center. The Microsoft Learn Az-900 learning path covers all exam objectives.

Q29 4 What is the Az-900 exam format?

Answer : Az-900 exam details: 40-60 questions (multiple choice, drag-and-drop, matching, case studies). Passing score: 700 out of 1000. Duration: 65 minutes. Cost: \$165 USD (varies by country). Available in: English, Japanese, Chinese, Korean, Spanish, Portuguese, French, German, Indonesian, Arabic, Russian. Can be taken online (proctored) or at a testing center. No expiration — Az-900 certification does not require renewal.

Q29 5 What are the Az-900 exam skill areas?

Answer : Az-900 covers four skill areas: Describe cloud concepts (25-30%): cloud models, benefits, service types. Describe Azure architecture and services (35-40%): compute, networking, storage, identity, Azure well-architected framework. Describe Azure management and governance (30-35%): cost management, Azure governance tools, monitoring, compliance. Study these weighted areas accordingly — Azure architecture and services has the most questions.

Q29
6 What is Microsoft Learn for Az-900?

Answer : Microsoft Learn (learn.microsoft.com) provides free, official, self-paced learning paths for Az-900. The Az-900 learning path has 3 modules: Microsoft Azure Fundamentals: Describe cloud concepts, Describe Azure architecture and services, Describe Azure management and governance. Each module contains interactive lessons, knowledge checks, and sandbox exercises in live Azure environments. Completing the learning path earns a badge and prepares for the official exam.

Q29
7 What are common Az-900 exam tips?

Answer : Key tips: Focus on understanding CONCEPTS not memorizing specifics. Know the difference between IaaS/PaaS/SaaS and which Azure services fall into each. Understand shared responsibility model thoroughly. Know CapEx vs OpEx, pricing calculator vs TCO calculator. Understand Azure's management hierarchy (Management Group > Subscription > Resource Group > Resource). Know availability zones vs availability sets vs regions. Practice with Microsoft Learn sandbox and official practice assessments.

Q29
8 What is the difference between Azure public cloud services and Azure Stack?

Answer : Azure (public cloud): Microsoft's global multi-tenant cloud infrastructure. Azure Stack: extends Azure to on-premises or edge locations. Azure Stack Hub: full Azure environment on-premises (regulatory requirements, disconnected scenarios). Azure Stack HCI: hyperconverged infrastructure for hybrid workloads. Azure Stack Edge: hardware appliance for AI processing at the edge. All Azure Stack products use the same Azure services, APIs, and tools as public Azure.

Q29
9 What are the key differences between Azure AD Free, P1, and P2?

Answer : Azure AD Free: user and group management, SSO to Microsoft 365 and SaaS apps, basic reports, Self-Service Password Reset for cloud users. Azure AD Premium P1 (included with Microsoft 365 Business Premium/E3): Conditional Access, hybrid identity (write-back), dynamic groups, self-service group management, MFA, Identity Protection (basic). Azure AD Premium P2 (included with Microsoft 365 E5): full Identity Protection (risk-based CA), Privileged Identity Management (PIM), access reviews.

Q30
0

What is the Azure Well-Architected Framework for Az-900?

Answer
:

The Well-Architected Framework's five pillars — Reliability, Security, Cost Optimization, Operational Excellence, Performance Efficiency — are tested at a conceptual level in Az-900. You should know: each pillar's purpose, that Azure Advisor provides recommendations aligned to these pillars, and that the Well-Architected Review helps assess workloads. Az-900 doesn't test implementation details — just the framework's purpose and Azure tools that support it.

Q30
1

What should you know about Azure pricing for Az-900?

Answer
:

Az-900 pricing key concepts: CapEx vs OpEx model shift. Consumption-based pricing (pay-as-you-go). Factors affecting cost: resource type, usage (meters), region, bandwidth. Ways to save: Reserved Instances, Spot VMs, Azure Hybrid Benefit, right-sizing. Tools: Pricing Calculator (estimate costs), TCO Calculator (compare on-premises vs Azure), Azure Cost Management (monitor actual spending), Azure Advisor (cost recommendations). Free account vs free tier services.